# EMT TentCare System Guide




**EMT TENTCARE**

*Patient flow for a six-tent emergency field hospital*

The complete guide: how the system works, the roles, and every screen and field

Demo version · October 2026

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Live demo: [mahassira.github.io/emergency-field-hospital](mahassira.github.io/emergency-field-hospital)

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> **Important:** this guide describes the demo version of the system. Every patient, staff member and number in it is fictional. The demo must not be used with real patient data. The real version is designed to run inside the organisation’s own Microsoft 365 environment (see chapters 15 and 16).

# 1. About the system

## 1.1 What EMT TentCare is

EMT TentCare is a patient flow system for a field hospital of six tents, which can grow, run by Emergency Medical Teams [10]. Its tagline is Know every patient's status. The idea is simple: one file per patient, opened at registration, that follows the patient from tent to tent until they leave. Each tent adds what it did: vital signs, the clinical assessment, lab tests and imaging, referrals, admission to the ward, and the visit outcome. Every step is recorded with the staff member's name and the time.

## 1.2 Why it was built

The system comes from working with the International Medical Corps (IMC) field hospital in Gaza, which first opened in January 2024 [5]. In the first days of any field hospital, the same problems keep coming back:

- Paper files get lost between tents, or the same details are written down in several places.
- Doctors don't know whether a test result is back, so someone is sent to ask.
- Nobody can quickly say how many ward beds are free.
- Management needs daily numbers for donors and the Health Cluster, and collects them by hand.
- The internet drops often, so any system that needs a constant connection stops.

EMT TentCare tries to solve these problems in one simple system that works on phones, tablets and computers, and keeps working when the internet is down.

Looking ahead, field hospitals will keep being set up in emergencies and conflict zones, often within days and in places where the internet is weak or cut off. They need a system that is ready to use from the first day, works both online and offline, and is prepared before the crisis rather than built during it. A ready system also gives donors and management the reporting they require, such as daily patient numbers, outcomes and the WHO EMT Minimum Data Set [9], without extra work for clinical staff. It also brings:

- **Faster start:** staff are trained before deployment and use the same screens at every site.
- **Continuity of care:** one record follows the patient across tents, referrals and return visits.
- **Accountability:** every action is recorded with a name and a time, which supports audits and donor reports.
- **Patient safety:** WHO-based triage criteria and checklists reduce missed steps when the team is under pressure.
- **Data protection:** patient data stays inside the organisation's own Microsoft 365 environment.
- **Reuse:** any Emergency Medical Team can deploy it with small changes, instead of starting from paper each time.

## 1.3 Who this guide is for

*Table 1: Where each reader should start*

| Reader | What to read |
|---|---|
| Management | Chapters 1, 2, 3, 11, 15, 16 and 21: the idea, the roles, the dashboard, security and the plan. |
| Clinical and admin staff | Chapters 4 to 10 and chapter 18: how each person uses their screens, step by step. |
| IT team | Chapters 14, 15, 16 and 22: offline use, security, and the move to Power Apps and Dataverse. |
| Monitoring and evaluation | Chapters 11, 13 and 22: indicators, reports and data quality. |

## 1.4 Where the project stands

- **Demo version:** live now at https://mahassira.github.io/emergency-field-hospital/, with fictional data. It saves everything on the device only.
- **Proposed real version** built inside the organisation's own Microsoft 365 environment, with sign-in by work account, a Microsoft Dataverse database and Power BI reports.
- **Code and documentation:** on GitHub in the Mahassira/emergency-field-hospital repository.

## 1.5 Based on the WHO Emergency Care Toolkit

The clinical forms follow the WHO Emergency Care Toolkit, which WHO publishes for free use: the Acute Referral Form, Acute Transfer Checklist, Counter-Referral Form and the trauma Emergency Unit Form (triage criteria and ABCDE). EMT TentCare is not an official WHO tool and is not endorsed by WHO or International Medical Corps; the organisation's clinical leads should review these forms before any real use [1][2][3].

# 2. The hospital, the tents and the patient journey

## 2.1 The six tents

The hospital has six numbered tents, plus a Lab & X-ray unit that serves all of them. Other tents, such as GBV, Child Protection and Logistics/Support, are not included yet.

*Table 2: The six tents*

| No. | Name | What happens there |
|---|---|---|
| 1 | Registration & Triage | The patient arrives, gets a file, has vital signs taken, is given a triage colour, and is sent to the right tent. |
| 2 | Emergency | Critical and urgent cases: injuries, breathing problems, acute medical cases. |
| 3 | Outpatient Clinics | Non-urgent cases: chronic illness, children, follow-up. |
| 4 | Maternity & Delivery | Pregnancy care, deliveries and postnatal care. |
| 5 | Operating Theatre | Surgery and major procedures. |
| 6 | Inpatient Ward | 50 beds for patients who need to stay. |
| — | Lab & X-ray | Serves every tent: receives test and imaging requests and sends results straight to the patient's file. |

Tents 2, 3, 4 and 5 are the **clinical tents**. These are the tents a patient can be sent to from registration or by referral.

In WHO terms, this hospital matches an **Emergency Medical Team Type 2**: inpatient acute care, general and obstetric surgery, at least 20 beds (here 50), one operating theatre, and open 24 hours a day [10]. With an intensive care unit and a second operating table, the same system can serve a **Type 3** referral hospital. **Type 1** teams, which give outpatient care only, can use the registration, triage, outpatient and follow-up parts.

## 2.2 The patient journey

This is the full path of a typical patient through the system:

1. **Arrival and registration (Tent 1):** the registration clerk enters the patient's details, allergies and vital signs, ticks any WHO triage criteria, picks a triage colour and chooses the tent. If the patient was here before, the system offers to open a new visit on the same file.
2. **File number and wristband:** the patient gets a file number straight away (for example FH-0026), printed on a wristband.
3. **Waiting in the tent queue:** the patient appears in the Waiting list of the chosen tent, sorted by triage colour and then waiting time.
4. **Consultation:** the doctor presses Start consultation. The patient moves to the In care list and the doctor is recorded as the assigned doctor.
5. **Assessment:** the doctor records the primary survey (ABCDE) for emergencies, then the history, examination and diagnosis. Medication and procedures are recorded with the time they were given, and the patient is reassessed as needed.

6. **Lab and imaging (if needed):** the doctor requests a test or X-ray. It appears at once in the lab. The technician enters the result and it shows in the patient's file.
7. **Referral (if needed):** if the patient needs another tent, such as the theatre, the doctor refers them. The receiving tent presses Accept patient, so nobody is lost between tents. If the patient needs another hospital, the doctor fills the external referral form and transfer checklist.
8. **Ward admission (if needed):** the doctor picks a free bed. The bed is reserved until the ward nurse confirms the patient has arrived.
9. **Discharge:** the doctor approves the discharge and records the visit outcome, take-home medication, any referral and a follow-up date. The patient counts as gone only when the departure is confirmed: by the nurse on the ward, or by the doctor in the other tents. A ward bed then goes to cleaning.
10. **Follow-up:** if a follow-up date was set, the patient appears in the follow-up list on that day and is marked as attended or missed.

## 2.3 Patient status

At any moment, a patient has one of four statuses:

*Table 3: Patient statuses*

| Status | Meaning | Where it shows |
|---|---|---|
| Waiting | In a tent, not yet seen. | The tent's Waiting list. |
| In care | Being examined or treated in the tent. | The tent's In care list. |
| Admitted (Bed N) | Admitted to the ward on a given bed. | The bed map in Tent 6. |
| Discharged | Left the hospital, was transferred, left without being seen, left against medical advice, died, or was dead on arrival. | Search and the patient's record only. |

## 2.4 File number

Every patient gets a unique file number in the form **FH-0001**, then FH-0002 and so on (FH stands for Field Hospital). It is printed on the wristband and any staff member can search by it. The number stays the same for the whole visit, however many tents the patient passes through. When a patient comes back after being discharged, they keep the same file number and get a new visit: FH-0004-V1 for the first visit, FH-0004-V2 for the second, and so on. Earlier visits stay in the file under Previous visits.

# 3. Roles and permissions

The system has five roles. Each role sees only the screens it needs and can only take actions that fit its job.

*Table 4: Roles and demo accounts*

| Role | Demo account | Person in the demo data | Screens |
|---|---|---|---|
| Registration clerk | reception@emttentcare.com | Reem Al-Masri | Register patient, Today's patients, Attendance, Search |
| Physician | doctor@emttentcare.com | Dr. Khaled Nasser | Emergency, Outpatient Clinics, Maternity & Delivery, Operating Theatre, Ward, Attendance, Search |
| Lab & X-ray technician | lab@emttentcare.com | Tariq Younis | Lab & X-ray, Attendance, Search |
| Ward nurse | nurse@emttentcare.com | Huda Saleem | Ward, Attendance, Search |
| Administrator (site manager) | admin@emttentcare.com | Site Manager | Every screen, plus the Dashboard |

## 3.1 Permissions table

✓ means the role can take the action. — means it cannot.

*Table 5: Permissions by role*

| Action | Registration | Physician | Lab | Nurse | Admin |
|---|---|---|---|---|---|
| Register a new patient | ✓ | — | — | — | ✓ |
| See the clinical tent queues | — | ✓ | — | — | ✓ |
| Start consultation, primary survey, clinical assessment | — | ✓ (clinical tents and ward) | — | — | ✓ |
| Request a test or X-ray | — | ✓ | — | ✓ (ward) | ✓ |
| Enter a test or X-ray result | — | — | ✓ | — | ✓ |
| Refer to another tent | — | ✓ | — | — | ✓ |
| Admit to the ward | — | ✓ | — | — | ✓ |
| Approve a discharge | — | ✓ | — | — | ✓ |
| Confirm a patient has left | — | ✓ | — | ✓ (ward) | ✓ |
| Update vitals, give medication, reassess | — | ✓ | — | ✓ (ward) | ✓ |
| Accept a patient sent from another tent | — | ✓ | — | ✓ (ward arrivals) | ✓ |
| Refer to another facility, record arrival and counter-referral | — | ✓ | — | — | ✓ |
| Update allergies | ✓ | ✓ | — | ✓ | ✓ |

| Action | Registration | Physician | Lab | Nurse | Admin |
|---|---|---|---|---|---|
| Start mass casualty mode | ✓ | — | — | — | ✓ |
| Mark a ward bed as cleaned and ready | — | — | — | ✓ | ✓ |
| View and print a wristband | ✓ | ✓ | ✓ | ✓ | ✓ |
| Check themselves in and out | ✓ | ✓ | ✓ | ✓ | ✓ |
| Change other staff members' attendance | ✓ | — | — | — | ✓ |
| Mark a follow-up visit as attended or missed | ✓ | ✓ | — | ✓ | ✓ |
| Dashboard and Excel export | — | — | — | — | ✓ |
| Search for any patient and open their file | ✓ | ✓ | ✓ | ✓ | ✓ |

> When someone opens a patient file they are not allowed to work on (for example the registration clerk or the lab technician), they see the whole file read-only. The action buttons are hidden and only the Wristband button shows. Once a patient is discharged, their file is read-only for everyone.

## 3.2 Each role in detail

**Registration clerk**

- Receives the patient in Tent 1, opens a file (or a new visit for a returning patient), takes vital signs and allergies, sets the triage colour with the WHO criteria and sends the patient to the right tent.
- Prints the wristband with the file number.
- Sees Today's patients: everyone registered today and where they are now. Useful when relatives ask.
- Manages staff attendance at the desk: check-in and check-out, absences and leave.
- Starts mass casualty mode when many patients arrive at once (section 5.7).

**Physician**

- Sees the queues of the four clinical tents and the ward.
- Starts consultations, records the primary survey (ABCDE) and the clinical assessment, gives and records medication, reassesses, and requests tests and imaging.
- Refers patients to another tent or another facility, admits them to the ward, and approves discharges.
- When they start a consultation or save the first assessment, they are recorded as the patient's assigned doctor.

**Lab & X-ray technician**

- Sees every pending request from every tent, sorted by how serious the patient is and then by request time.

- Enters the result, which goes straight to the patient’s file. The doctor sees “Results ready”.
- Sees the last 10 completed results.

**Ward nurse**

- Sees the map of the 50 beds: occupied beds with the patient’s name and triage colour, and free beds.
- Works on ward patients: vital signs, nursing notes, medication given, reassessment and test requests. Confirms when a new patient arrives on the ward.
- Confirms when a discharged patient has actually left. The bed then shows Cleaning until the nurse marks it ready.

**Administrator (site manager)**

- Sees everything and can take every action.
- The Dashboard gives a live picture of the whole hospital.
- Manages attendance for all staff and exports all data to Excel.

# 4. Signing in and getting around

## 4.1 Sign-in page

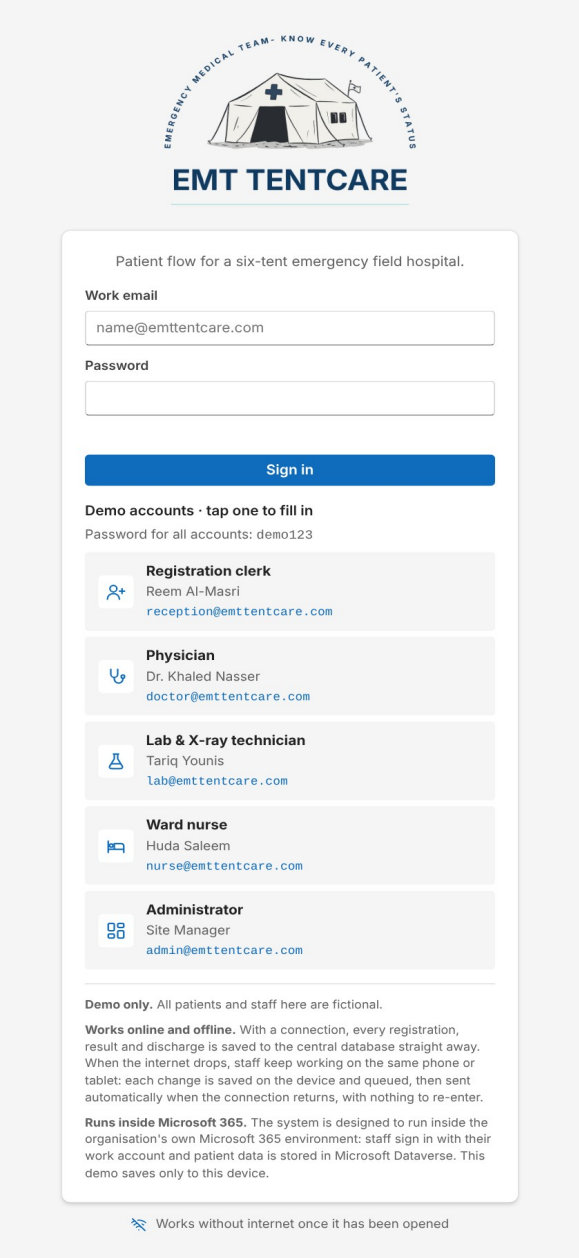



*Figure 1: Sign-in page on a computer*

1. Open the system's link in a browser.
2. Type your work email in **Work email** and your password in **Password**.
3. Press **Sign in**.

In the demo, a list of **Demo accounts** sits under the form. Tapping one fills in the email and password. You can also type just the short name, such as reception instead of reception@emttentcare.com.

At the bottom of the sign-in page, three short paragraphs explain that this is a demo, how the system works online and offline, and that it is designed to run inside Microsoft 365.

A wrong email or password shows “Wrong email or password”. After signing in, the user stays signed in until they press Sign out, even if the browser is closed.

## 4.2 On a computer

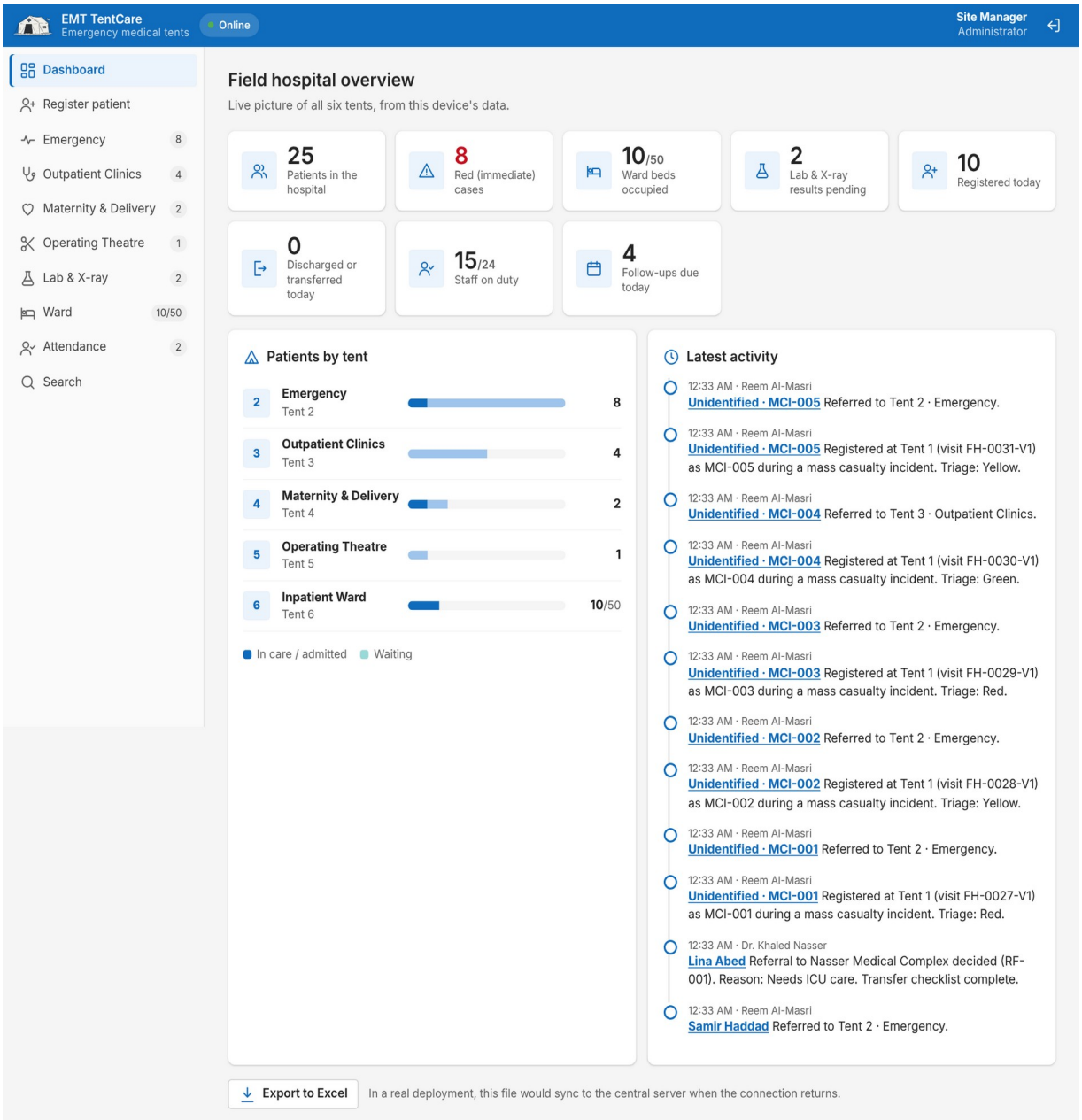



*Figure 2: The site manager’s home screen on a computer*

- **Blue top bar:** the logo and name, the connection indicator (Online or Offline), the user’s name and role, and the sign-out button.
- **Side menu:** from 900 pixels wide, the menu sits on the left. On smaller screens it shows as tabs under the top bar.
- **Numbers next to tent names:** how many patients are in each tent right now. Next to Ward, the number of occupied beds out of 50. Next to Attendance, the number of absent staff.
- **Bottom of each page:** a reminder that the data is fictional, and a Reset demo data link.

## 4.3 On a phone

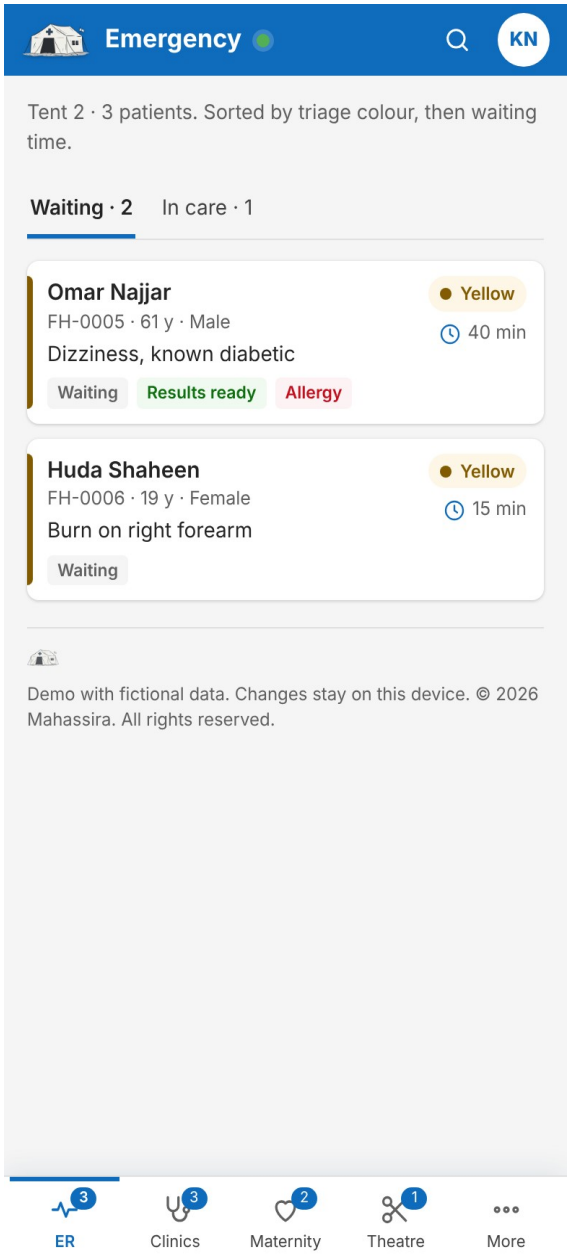



*Figure 3: Emergency queue on a phone*

- **Top bar:** the title of the current screen, a coloured dot for the connection (green online, red offline), the search button and the user's initials.
- **Tapping the initials** opens the account menu: check in or out of the shift, go to Attendance, reset the demo data, and sign out.
- **Bottom bar:** the role's main screens, and a **More** button for the rest when there are more than four.
- **Inside a patient file:** a back button at the top, and at the bottom a bar with the main action and an **Actions** button for the rest.

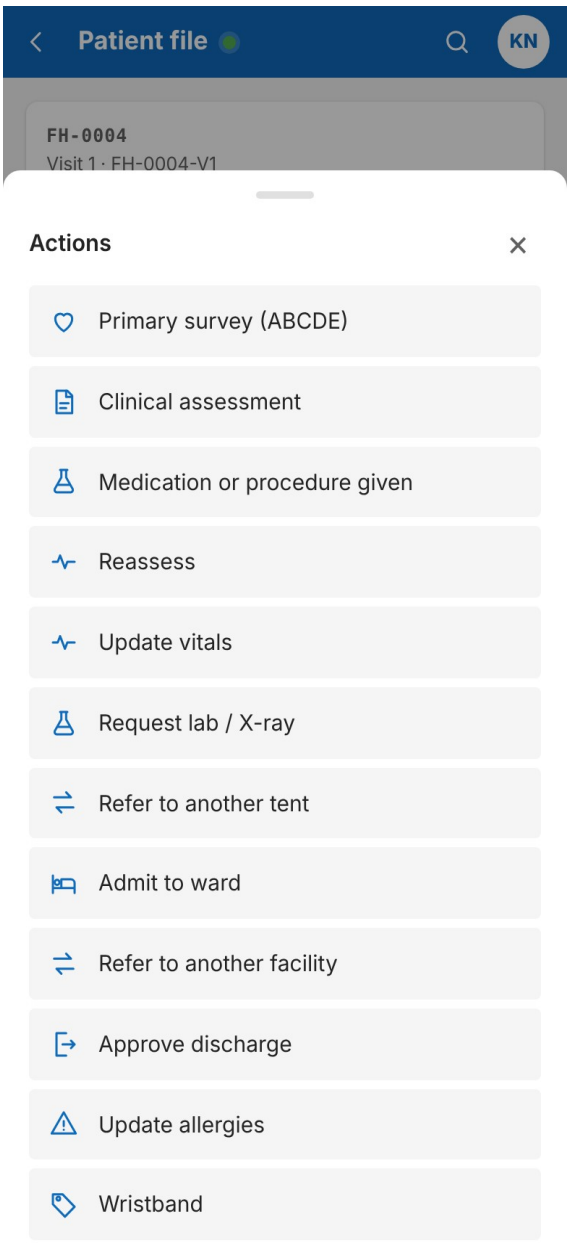



*Figure 4: The actions menu in a patient file on a phone*

# 5. Registration and triage (Tent 1)

The **Register patient** screen is the first screen the registration clerk sees. This is where a patient's file is opened and the patient is sent to the right tent.

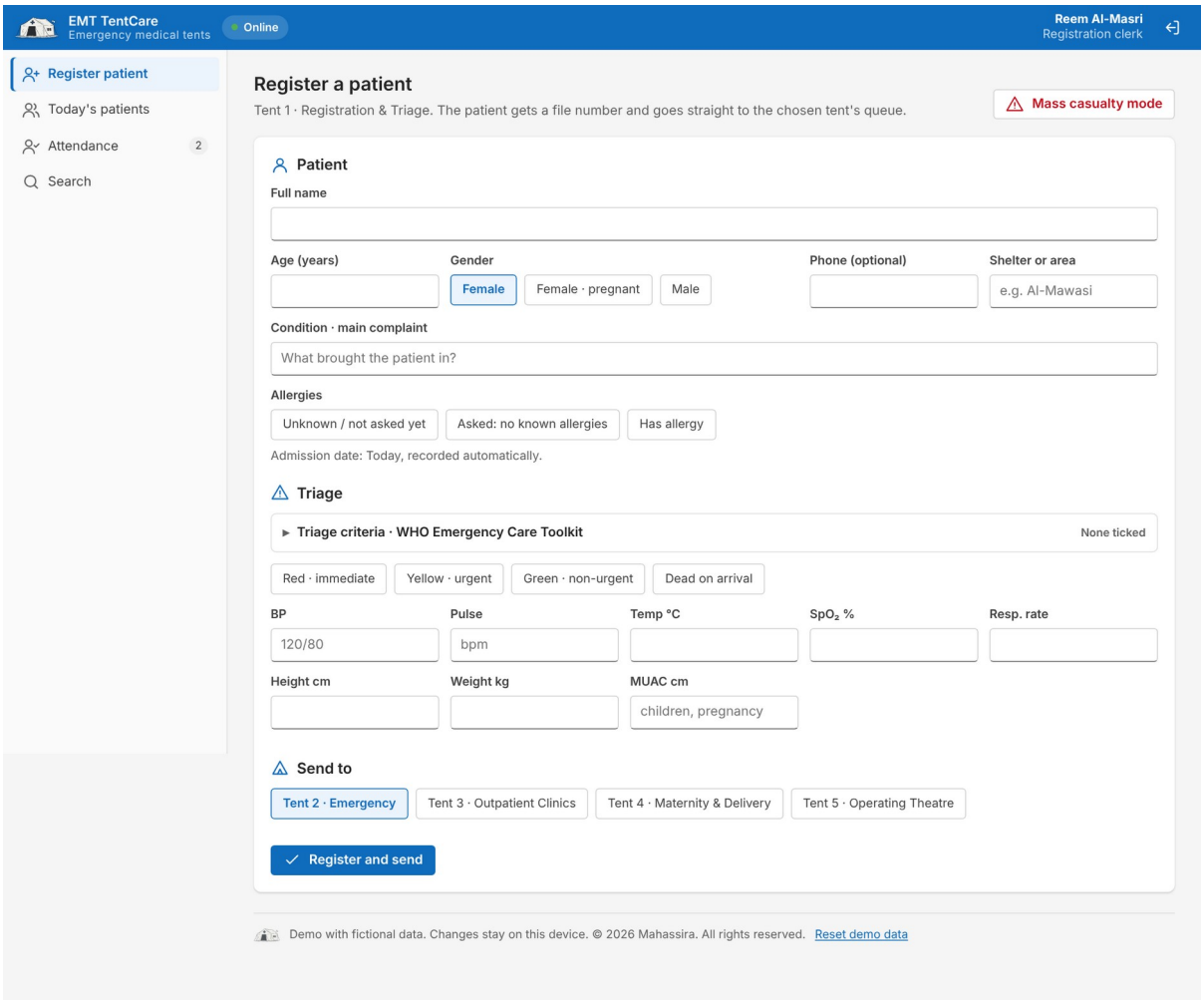



*Figure 5: The new patient form*

## 5.1 Form fields

*Table 6: Registration form fields*

| Field | Required? | Notes |
|---|---|---|
| Full name | Yes | The patient's name as they give it. |
| Age (years) | Yes | 0 to 120. |
| Gender | Yes | Three options: Female, Female · pregnant, Male. The pregnant option puts a "Pregnant" badge on her file everywhere. |
| Phone | No | Optional, for follow-up contact. |
| Shelter or area | No | For example Al-Mawasi or Khan Younis. |
| Condition · main complaint | Yes | Why the patient came, in simple words. |

| Field | Required? | Notes |
|---|---|---|
| Allergies | Yes | Three choices, none selected at first: Unknown / not asked yet, Asked: no known allergies, Has allergy. Has allergy needs the allergy written. Unknown shows an amber “Allergies unknown” bar in the file until someone asks. |
| Admission date | Automatic | Recorded automatically when the form is saved. |
| Triage | Yes | Red, yellow, green, or dead on arrival. Nothing is selected at first, so the clerk must choose. See 5.2 and 5.5. |
| BP | No | In the form 120/80. Impossible values are refused, and unusual values ask for a second press (see 5.1.1). |
| Pulse | No | Beats per minute (bpm). |
| Temp °C | No | Degrees Celsius, with one decimal. |
| $SpO_2$ % | No | Oxygen saturation, as a percentage. |
| Resp. rate | No | Breaths per minute. |
| Height cm | No | In centimetres. |
| Weight kg | No | In kilograms. |
| MUAC cm | No | Mid-upper arm circumference, in centimetres. Used to screen children and pregnant women for malnutrition. |
| Send to | Yes | One of the four clinical tents: Emergency, Outpatient Clinics, Maternity & Delivery, Operating Theatre. |

### 5.1.1 Checks on the vital signs

The form refuses values that cannot be real, and asks you to look again at values that are possible but unusual. For an unusual value, a message explains it and a second press of the button saves anyway.

*Table 7: Limits for vital signs*

| Sign | Refused (impossible) | Asks to re-check (unusual) |
|---|---|---|
| BP | Not written like 120/80; systolic outside 50 to 260; diastolic outside 20 to 160 or not lower than systolic | Systolic under 80 or over 200; diastolic over 120 |
| Pulse | Outside 20 to 250 bpm | Under 40 or over 180 |
| Temperature | Outside 30 to 45 °C | Under 35 or over 40.5 °C |
| $SpO_2$ | Outside 30 to 100% | Under 85% |
| Resp. rate | Outside 4 to 80 per minute | Under 8 or over 40 |
| Height, weight, MUAC | Outside 30 to 250 cm, 0.5 to 300 kg, 5 to 50 cm | — |

## 5.2 Triage colours

*Table 8: Triage colours*

| Colour | Label in the system | Meaning | Queue priority |
|---|---|---|---|
| Red | Red · immediate | Critical, needs treatment now. | Always first |
| Yellow | Yellow · urgent | Urgent but stable for now. | After red |
| Green | Green · non-urgent | Not urgent. | Last |
| Black | Dead on arrival | The patient died before arrival. The file is recorded but the patient is not sent to a tent. | — |

The triage colour shows everywhere: a coloured bar on the patient card, a coloured badge next to the name, and a coloured dot on the patient's ward bed. Red, yellow and green follow the Interagency Integrated Triage Tool used by WHO, ICRC and MSF [11].

## 5.3 After pressing "Register and send"

1. The patient gets a new file number.
2. The patient's record shows "Registered at Tent 1. Triage: Red." with any ticked criteria, then "Referred to Tent 2 · Emergency."
3. The system opens the patient's file and shows the **wristband** with the file number.
4. The patient appears at once in the Waiting list of the chosen tent.

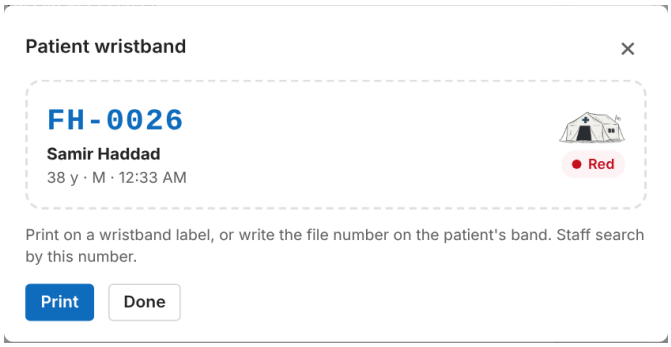



*Figure 6: The patient wristband*

The wristband shows the file number in large type, the name, age, sex, arrival time and triage colour. **Print** prints it on a wristband label, or staff can write the number on the band by hand. The wristband can be opened later from the patient file with the **Wristband** button.

## 5.4 Today’s patients

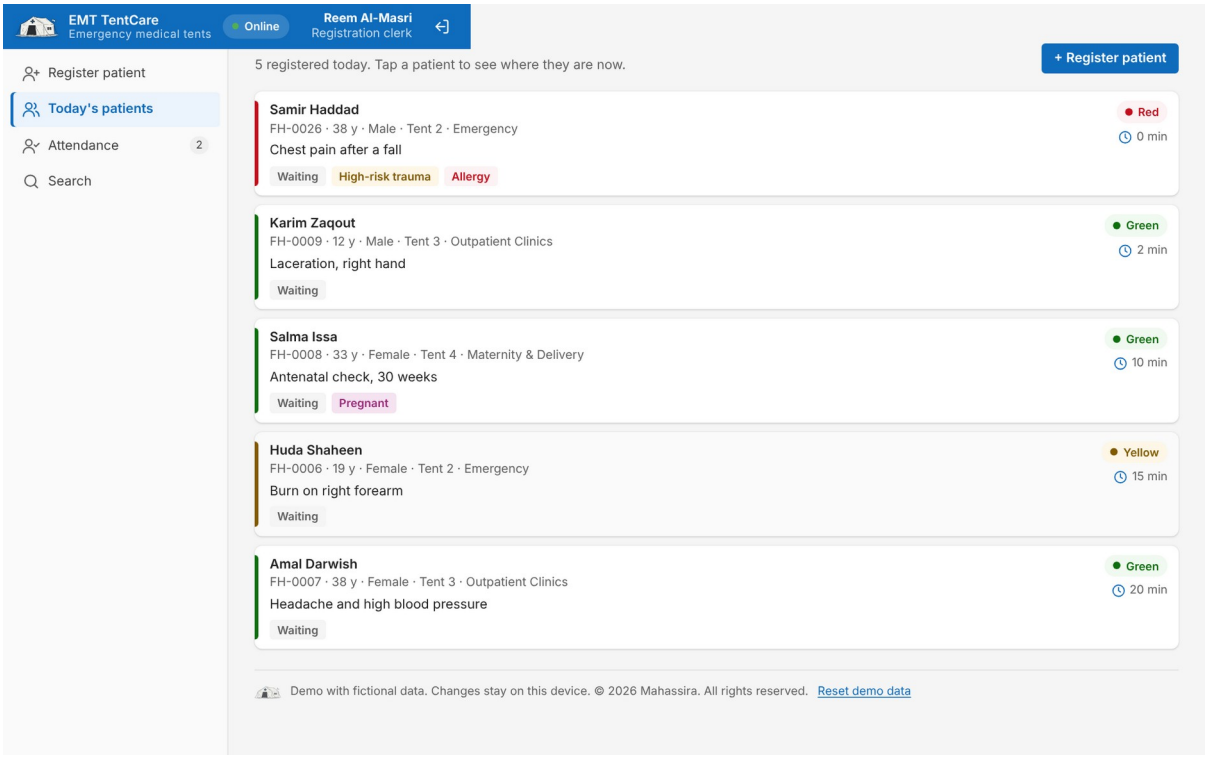



*Figure 7: Today’s patients*

The **Today’s patients** screen lists everyone registered today, newest first, with where each patient is now. It has a “+ Register patient” button for the next patient.

## 5.5 Triage criteria (WHO)

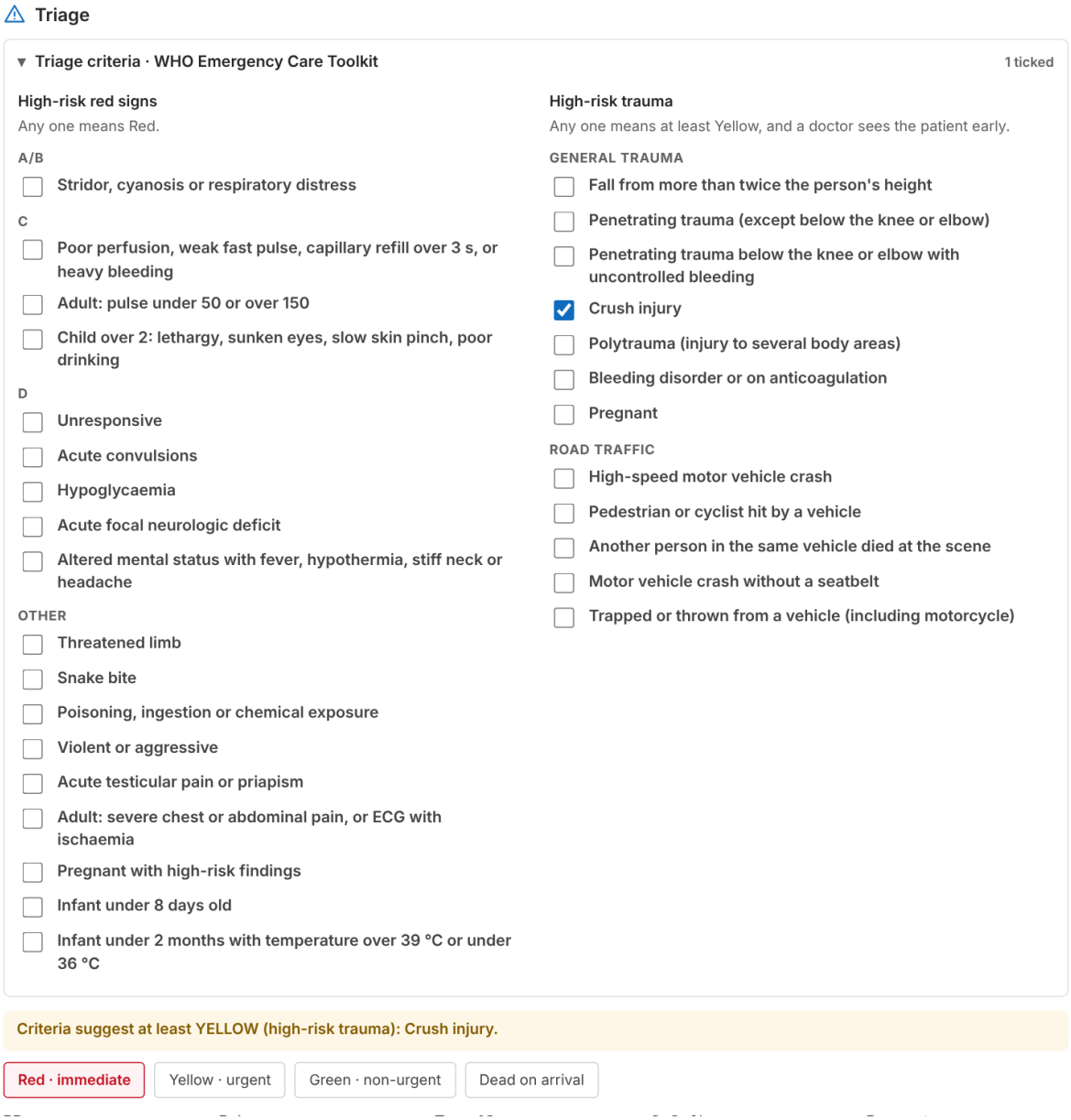



*Figure 8: Triage criteria ticked on the registration form*

Above the triage colours, a section called “Triage criteria · WHO Emergency Care Toolkit” opens with one tap. It lists the two groups from the WHO trauma form [3]:

- **High-risk red signs:** grouped as A/B (airway and breathing), C (circulation), D (disability) and Other, for example stridor, heavy bleeding, an adult pulse under 50 or over 150, convulsions, snake bite or poisoning. **Any one means Red.**
- **High-risk trauma:** general trauma (fall from more than twice the person’s height, penetrating trauma, crush injury, polytrauma, anticoagulation, pregnancy) and road traffic (high-speed crash, pedestrian or cyclist hit, trapped or thrown from a vehicle). **Any one means at least Yellow,** and a doctor should see the patient early.

A coloured line under the section shows what the ticked criteria suggest. If the clerk types an adult pulse under 50 or over 150, the pulse sign is ticked automatically.

> If the clerk picks a lower colour than the criteria suggest, a warning appears. Pressing again saves, and the file records that the colour was kept by clinical judgement, with the clerk's name. The criteria also show on the patient file and in the Excel export.

## 5.6 Returning patients

When the name, sex and age (within one year) match a patient already registered, a "Possible existing patient" window appears before anything is saved. For a discharged patient, Same person: start a new visit opens visit 2 on the same file. A known allergy is kept even if the desk chose Unknown this time. Different person: create a new file makes a new file number. A similar name alone is not proof it is the same person, so check with the patient.

## 5.7 Mass casualty mode

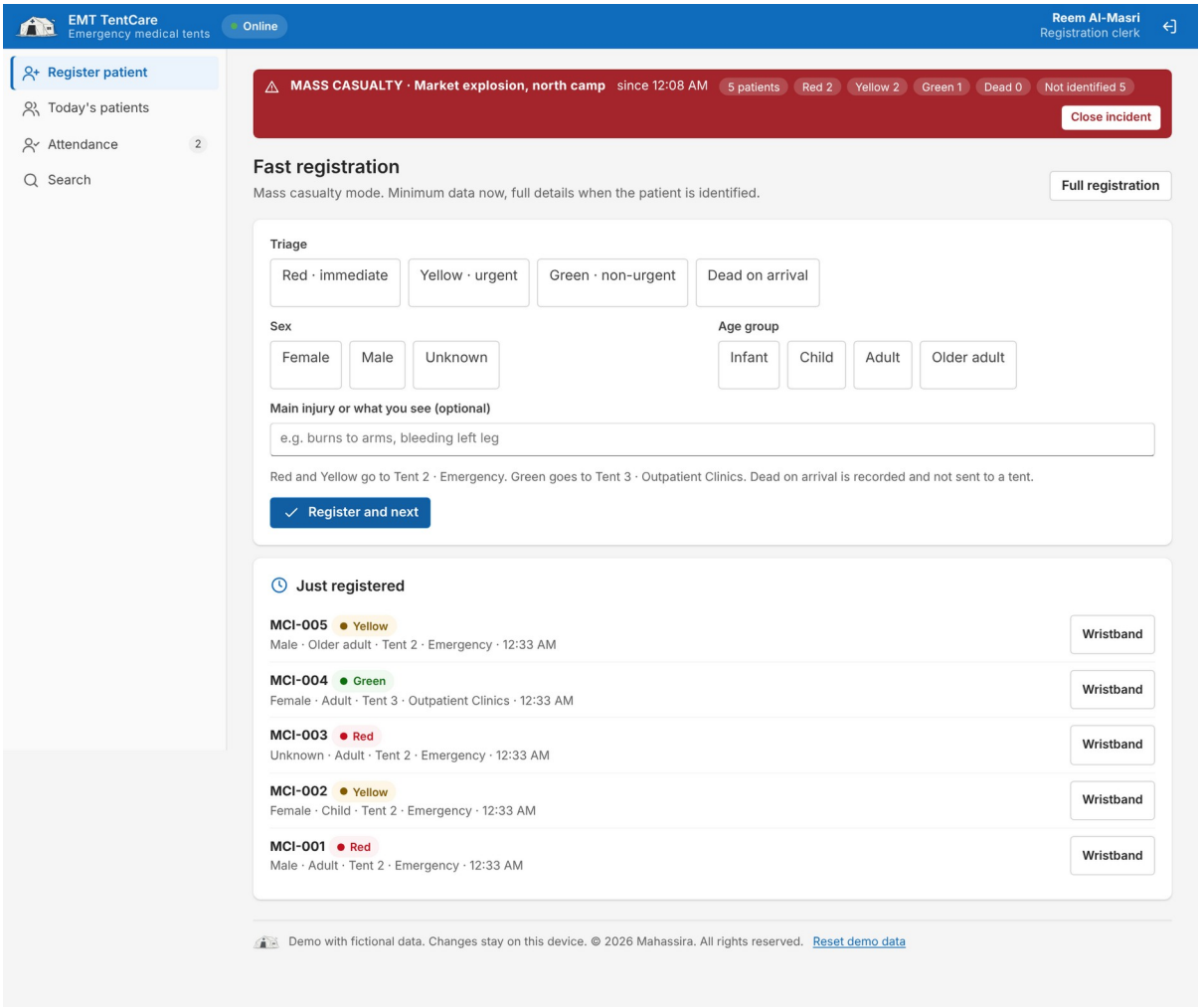



*Figure 9: Fast registration during a mass casualty incident*

Use this when many patients arrive at once. It follows WHO guidance on mass casualty management [13]. The registration clerk or site manager presses Mass casualty mode on the registration screen and writes a short name for the incident.

- **Red banner:** every user, on every screen, sees the incident name, how long it has been running, and the number of patients by colour and not yet identified.

- **Fast registration:** only three choices: triage colour (including dead on arrival), sex (or Unknown) and age group (Infant, Child, Adult, Older adult). The main injury is optional. Press Register and next for the next patient.
- **Temporary ID:** each patient gets MCI-001, MCI-002 and so on, on top of the normal file number. Staff can search by it, and the wristband shows it.
- **Routing:** Red and Yellow go to Tent 2 · Emergency, Green to Tent 3 · Outpatient Clinics. Dead on arrival is recorded and not sent to a tent.
- **Identify later:** in the patient file, Identify patient adds the real name, age, sex and contact. The file number stays the same and the temporary ID stays on record. If the name matches another file, a warning appears.
- **Close the incident:** from the banner. The system says how many patients are still not identified, then returns to normal registration. Full registration is also available during the incident.

# 6. Tent queues

Each clinical tent has a screen with two lists: **Waiting** and **In care**. On a computer the lists sit side by side. On a phone they are two tabs.

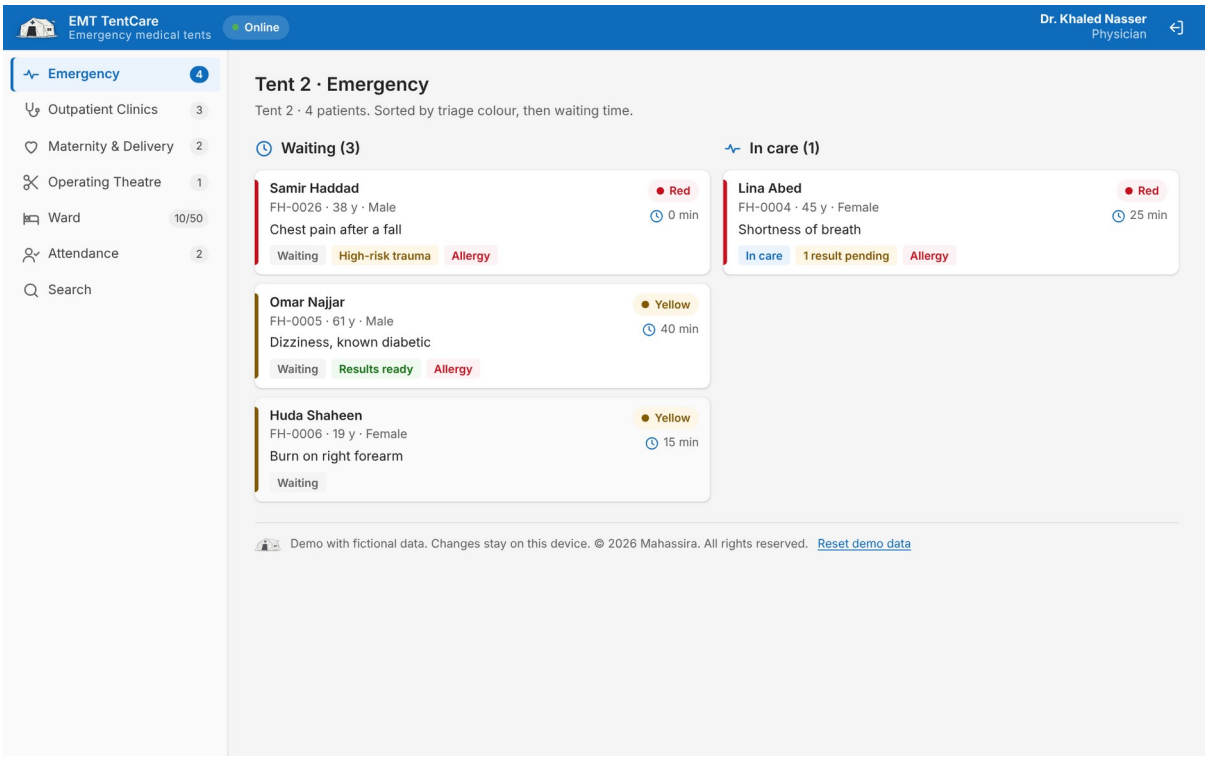



*Figure 10: The Emergency tent queue, as the doctor sees it*

## 6.1 Queue order

Patients are sorted automatically: **red first, then yellow, then green**. Within each colour, whoever has waited longest comes first. Waiting time counts from when the patient last moved into this tent, not from registration.

## 6.2 What a patient card shows

*Table 9: What a patient card shows*

| Item | Meaning |
|---|---|
| Coloured bar | The triage colour. |
| Name | The patient’s name. Tapping the card opens the file. |
| Second line | File number, age and sex. |
| Third line | Main complaint. |
| Colour badge and clock | Triage colour, and how long since the patient arrived in this tent. |
| Waiting / In care / Bed N | The patient’s status. |
| N result(s) pending | Tests or imaging still waiting for a result. |
| Results ready | New results came back since the patient arrived in this tent. |
| Pregnant | The patient is pregnant. |

| Item | Meaning |
|---|---|
| Allergy | The patient has a recorded allergy. |
| Not yet accepted / Bed reserved | Sent from another tent, or to a ward bed, and not yet received. |
| Discharge approved | The doctor approved the discharge; the patient has not left yet. |
| Visit N | A returning patient on their second or later visit. |
| Red sign / High-risk trauma | WHO triage criteria were ticked at registration. |
| Not identified | A mass casualty patient still known only by a temporary ID. |
| Referral out · not left | A referral to another facility is decided and the patient is still here. |

# 7. The patient file and every action on it

The patient file is the heart of the system. It holds everything that has happened to the patient since arrival, and every action is taken from it.

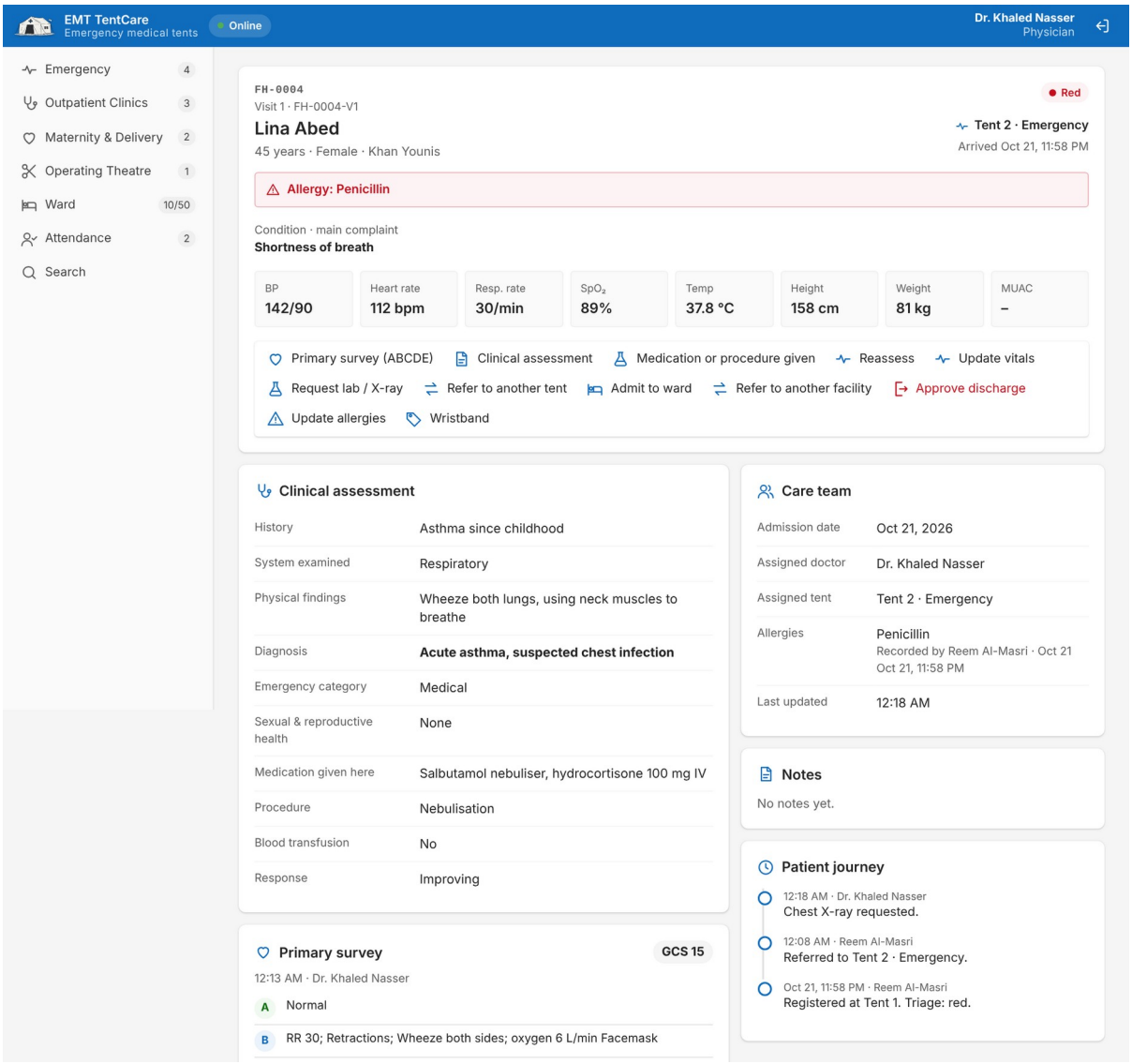



*Figure 11: The patient file on a computer*

## 7.1 Parts of the file

*Table 10: Parts of the patient file*

| Part | What it shows |
|---|---|
| Header | File number, visit number (for example FH-0004-V2), name, age, sex, shelter, phone, pregnancy badge, triage colour, current tent and bed, and arrival time. A blue line shows a transfer still waiting to be accepted. |
| Allergy bar | Red with the allergy, for example “Allergy: Penicillin”. Amber “Allergies unknown” when nobody has asked yet. Nothing when there are no known allergies. |
| Condition · main complaint | The main complaint. |

| Part | What it shows |
|---|---|
| Vital signs (8 boxes) | BP, Heart rate, Resp. rate, $SpO_2$, Temp, Height, Weight, MUAC. An empty box shows “–“. |
| Action buttons | Show only to people allowed to work on this patient. |
| Clinical assessment | History, examination, diagnosis and the rest of the assessment. |
| Primary survey | The latest ABCDE survey, one line per letter, with the GCS. |
| Medication & procedures given | Each drug or procedure with its time and who gave it. |
| Reassessments | Each reassessment: better, same or worse, vital signs, pain and any triage change. |
| Lab & imaging | Every test and imaging request, its status and its result. |
| Outcome & referral | Visit outcome, take-home medication, referral, notes, follow-up date, who approved the discharge, and who owns any pending results. Shows “Still in care” until discharge. |
| External referral | Shows when the patient is referred to another facility: the steps decided, left, arrived and counter-referral, and the receiving facility. |
| Previous visits | Earlier visits on the same file, with the diagnosis and outcome of each. |
| Care team | Admission date, assigned doctor, current tent, allergies, and when the file was last updated. |
| Notes | Doctors’ notes with time, name and tent number. |
| Patient journey | The full timeline: every action with its time and the staff member who took it. |

## 7.2 Actions

On a computer the actions are buttons. On a phone, the main action is a button at the bottom with an **Actions** button for the rest.

*Table 11: Actions in the patient file*

| Button | When it shows | What it does |
|---|---|---|
| Start consultation | When the patient is Waiting. | Moves the patient to In care and records the doctor as assigned. |
| Accept patient / Patient arrived on ward | When the patient was sent from another tent or admitted and not yet received. | Confirms the patient is here. Until then the other actions are limited. |
| Primary survey (ABCDE) | Doctor. | Records the A to E survey, FAST, and the injury mechanism. |
| Clinical assessment | Always, for allowed roles. | Opens the clinical assessment form. |
| Medication or procedure given | Doctor, and the nurse on the ward. | Records a drug with dose, route and time, or a procedure and time. |
| Reassess | Doctor, and the nurse on the ward. | Records better, same or worse, vital signs, AVPU, pain and the triage colour now. |
| Nursing note | Nurse on the ward. | Adds a short nursing note to the file. |

| Button | When it shows | What it does |
|---|---|---|
| Update vitals | Always, for allowed roles. | Updates the eight vital signs. |
| Request lab / X-ray | Always, for allowed roles. | Sends a test or imaging request to the lab. |
| Refer to another tent | Always, for allowed roles. | Moves the patient to another clinical tent's queue. |
| Admit to ward | When the patient is not already in the ward. | Admits the patient to a free bed. |
| Refer to another facility | Doctor. | Opens the WHO-based referral form and transfer checklist. |
| Approve discharge | Doctor, while the patient is in care. | Records the visit outcome and approves the discharge. The patient does not leave yet. |
| Confirm departure | After the discharge is approved: the nurse on the ward, or the doctor. | Records that the patient left. Frees the bed (it goes to cleaning) and creates the follow-up. |
| Update allergies | Everyone except the lab, while the patient is in the hospital. | Sets Unknown, No known allergies or the allergy; the change is recorded. |
| Identify patient | Mass casualty patients not yet identified. | Adds the real name, age, sex and contact. |
| Start new visit | Discharged patients; registration, doctor or admin. | Opens a new visit on the same file. |
| Wristband | For everyone. | Shows and prints the wristband. |

### 7.2.1 Clinical assessment

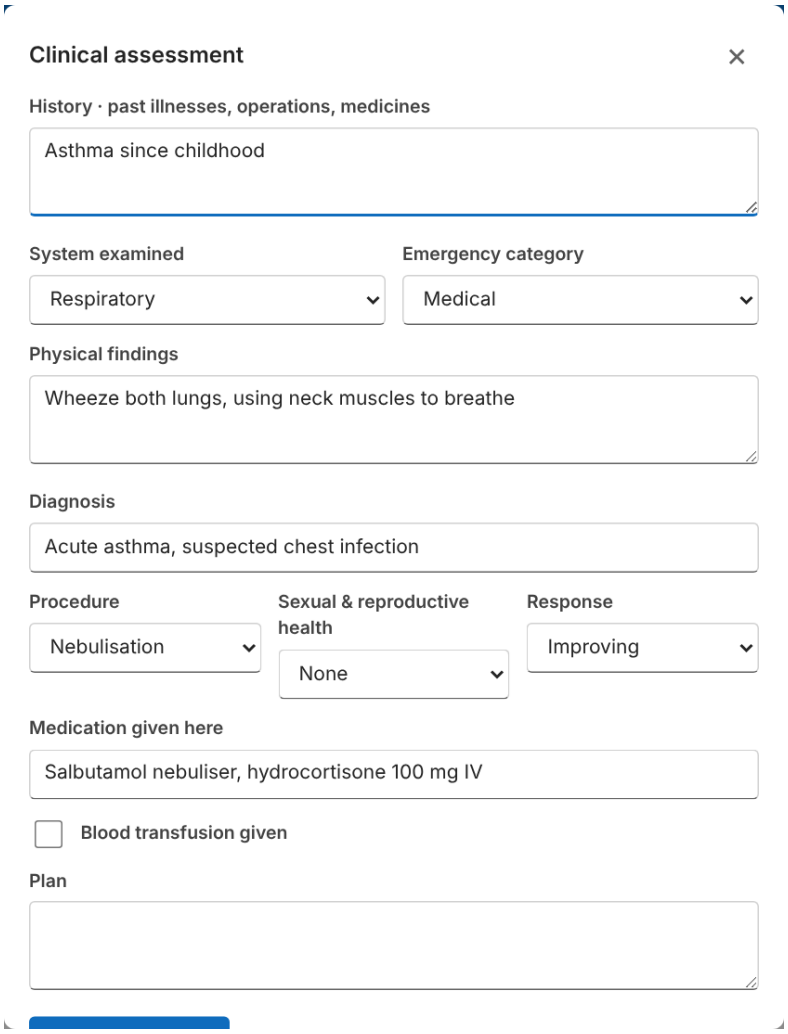



*Figure 12: The clinical assessment form*

*Table 12: Clinical assessment fields*

| Field | Options or notes |
|---|---|
| History | Past illnesses, operations and medicines. |
| System examined | General, Respiratory, Cardiovascular, Abdominal, Musculoskeletal, Neurological, Skin & wounds, Obstetric & gynaecological, ENT & eyes. |
| Emergency category | Trauma, Medical, Surgical, Obstetric, Paediatric, Burns, Mental health, Not an emergency. |
| Physical findings | Results of the physical examination. |
| Diagnosis | Required. |
| Procedure | None, Laceration repair, Wound dressing, Splint or cast, Fracture reduction, IV fluids, Incision and drainage, Nebulisation, Urinary catheter. |
| Sexual & reproductive health | None, Antenatal care, Delivery care, Postnatal care, Family planning, GBV clinical care (clinical care for survivors of gender-based violence). |
| Response | Improving, Healing, Stable, No change, Deteriorating. |
| Medication given here | Medicine given in the hospital: drug, dose and route. |
| Blood transfusion given | Whether the patient received blood. |

| Field | Options or notes |
|---|---|
| Plan | The treatment plan. Saved as a note in the Notes section. |

On save, the Clinical assessment section of the file updates, a note is added to Notes, and the record shows “Assessment: diagnosis (category)”. A waiting patient moves to In care, and if the patient has no assigned doctor, the current doctor is recorded. The form always opens filled with the latest assessment, so it can be edited.

### 7.2.2 Update vitals

A form with the eight vital signs, filled with the latest values. On save, the boxes in the file update and the record gets a line with all the values.

### 7.2.3 Request lab / X-ray

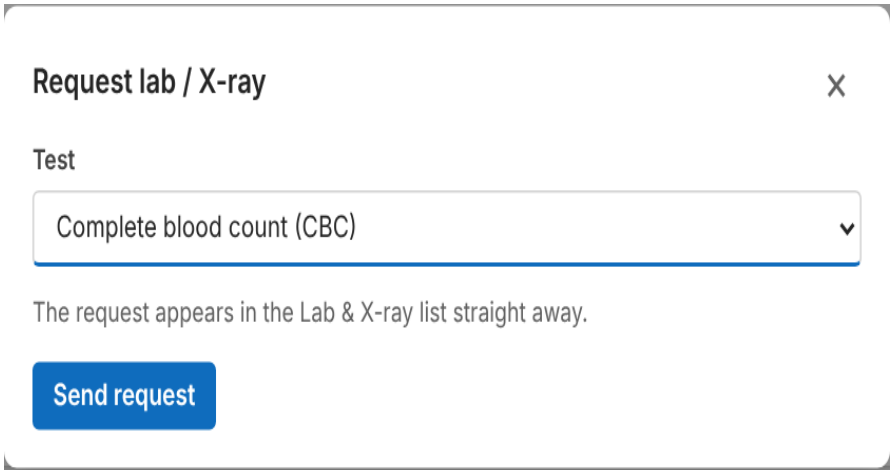



*Figure 13: Requesting a test or X-ray*

*Table 13: Available tests and imaging*

| Type | Available tests |
|---|---|
| Laboratory | Complete blood count (CBC), Blood glucose, Haemoglobin, Blood group & crossmatch, Urinalysis, Pregnancy test |
| Imaging | Chest X-ray, Limb X-ray, Abdominal ultrasound, Obstetric ultrasound |

The request appears at once on the lab screen, and in the patient file as “Pending” until the result comes back.

### 7.2.4 Refer to another tent

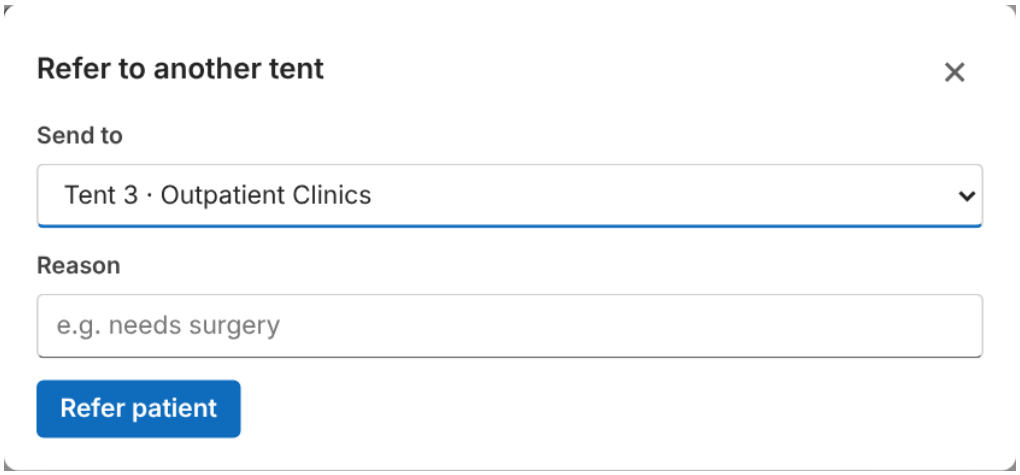



*Figure 14: Referring a patient*

Pick the clinical tent (the current tent is not in the list) and write the reason. The patient joins the new tent's queue with a "Not yet accepted" badge. A doctor there presses Accept patient. Until then, the sending doctor can press Cancel transfer to bring the patient back. If the patient was in the ward, their bed goes to cleaning.

### 7.2.5 Admit to ward

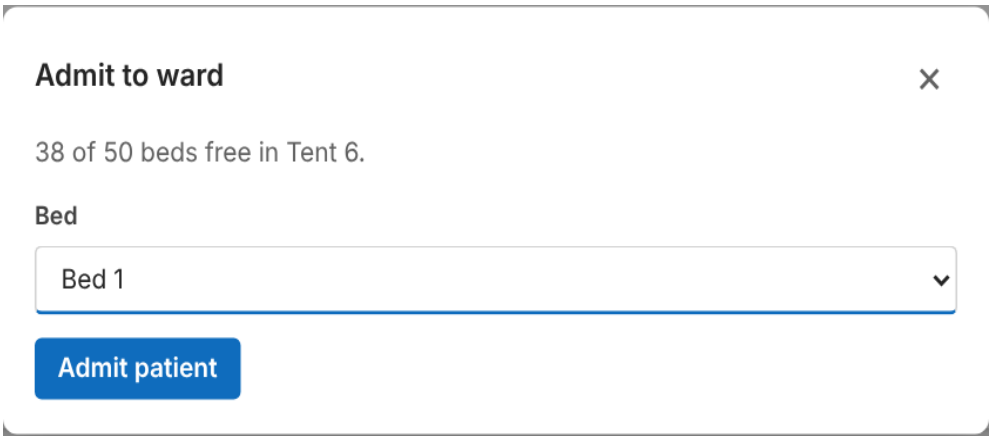



*Figure 15: Admitting a patient to the ward*

The form lists only beds that are free and clean. On save the bed is reserved and shows as Reserved on the ward map. When the patient arrives, the ward nurse (or a doctor) presses Patient arrived on ward. If all 50 beds are taken or being cleaned, a "Ward is full" message asks to discharge or transfer a patient first.

### 7.2.6 Approve discharge and confirm departure

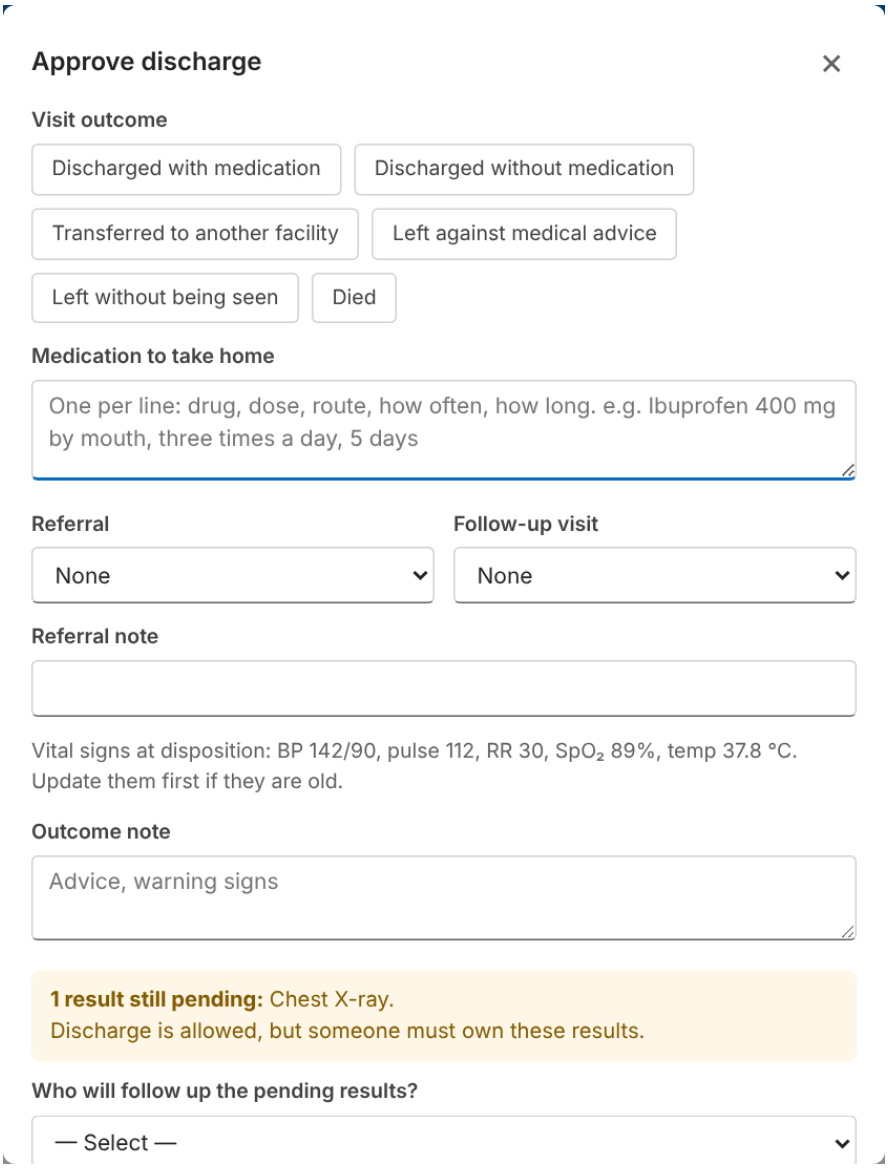



*Figure 16: Approving a discharge*

*Table 14: Discharge form fields*

| Field | Options or notes |
|---|---|
| Visit outcome | Required, nothing selected at first: Discharged with medication, Discharged without medication, Transferred to another facility, Left against medical advice, Left without being seen, Died. |
| Medication to take home | One per line: drug, dose, route, how often, how long. Required for “with medication”, and must be empty for “without medication”. |
| Referral | None, Diagnostic test, Specialist consultation, Another facility, Nutrition programme, Mental health support. |
| Follow-up visit | None, In 3 days, In 7 days, In 14 days. |
| Referral note | Details of the referral. |
| Outcome note | Advice and the warning signs to come back for. |
| Pending results owner | Shows only when a test is still pending. Choose the doctor who will follow up the result. |

Discharge has two steps, so a bed is never freed for a patient who is still there:

- **Approve discharge (doctor):** saves the outcome and shows “Discharge approved” on the file and card. The doctor can still cancel the approval.
- **Confirm departure:** on the ward, the nurse confirms the patient has left; in other tents the doctor does it, or presses “Approve and confirm the patient has left” in one step. The patient becomes Discharged, the bed goes to cleaning, and the follow-up visit is created (see chapter 10).

The form also shows the latest vital signs, which are saved as the vital signs at disposition.

### 7.2.7 Primary survey (ABCDE)

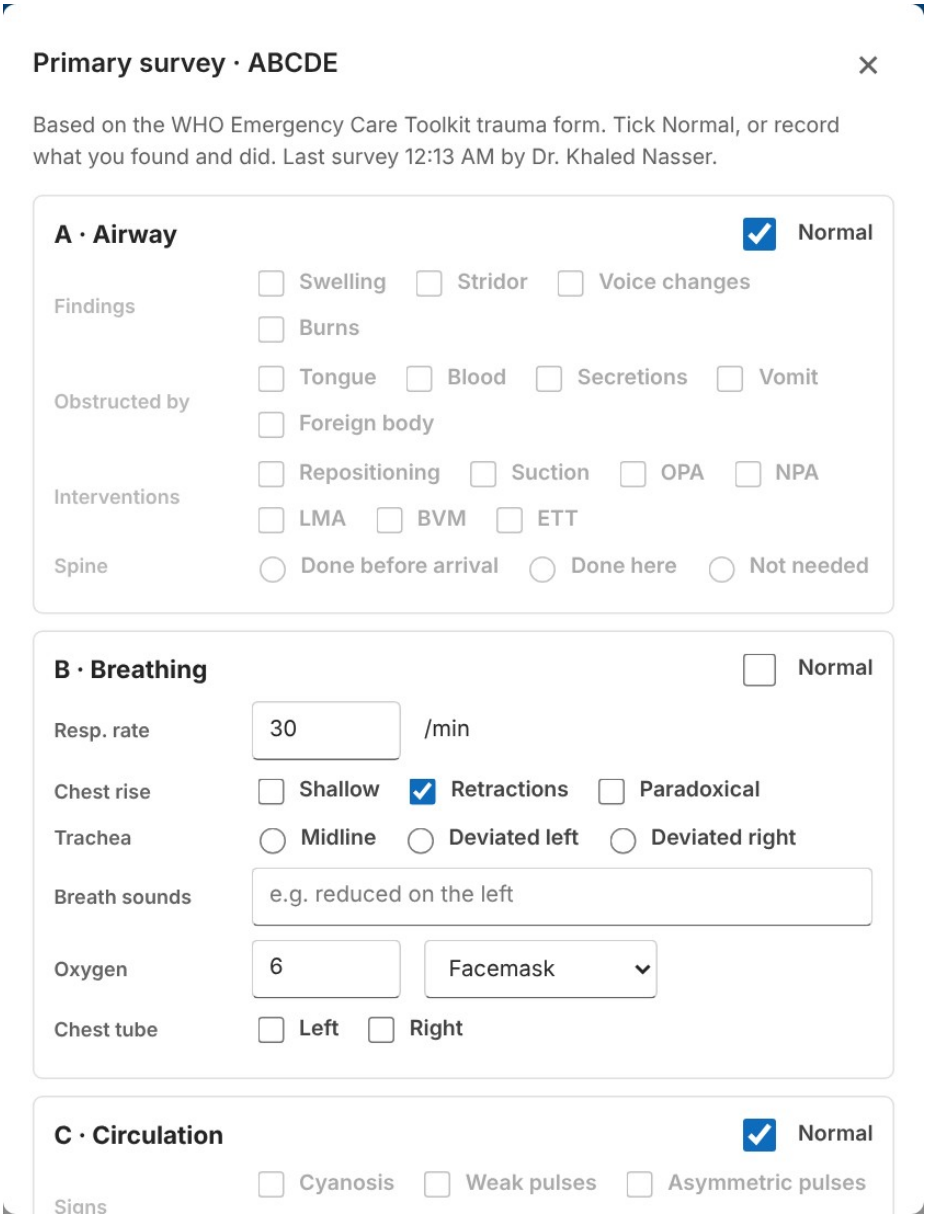



*Figure 17: The primary survey form*

Based on the WHO trauma form [3] and the WHO-ICRC Basic Emergency Care approach [12]. GCS is the Glasgow Coma Scale [14]. Each letter has a Normal box; untick it to record what you found and what you did.

*Table 15: Primary survey (ABCDE) fields*

| Letter | What is recorded | Checks |
|---|---|---|
| A · Airway | Swelling, stridor, voice changes, burns; what blocks the airway; interventions (suction, OPA, NPA, LMA, BVM, ETT); spine stabilisation. | Normal cannot be ticked together with a finding. |
| B · Breathing | Resp. rate, chest rise, trachea, breath sounds, oxygen (L/min and device), chest tube. | Resp. rate 0 to 80. |
| C · Circulation | Cyanosis, pulses, JVP, unstable pelvis, skin, capillary refill, bleeding control, access, fluids, blood, pelvic binder. | — |
| D · Disability | AVPU, GCS (E, V, M, added up as you type), limb movement, pupils, glucose, interventions. | GCS 8 or less, and glucose under 65 mg/dL, give a warning. |
| E · Exposure | Fully exposed and examined; wounds, burns, deformities. | — |
| F · FAST | Done, not indicated or not available; abdomen and chest results. | A positive FAST gives a warning. |

Mechanism of injury and intent (unintentional, self-harm, assault, conflict) are recorded at the end. Each letter must be ticked Normal or have something recorded. The latest survey shows in the file and fills the ABCDE part of an external referral.

### 7.2.8 Medication or procedure given

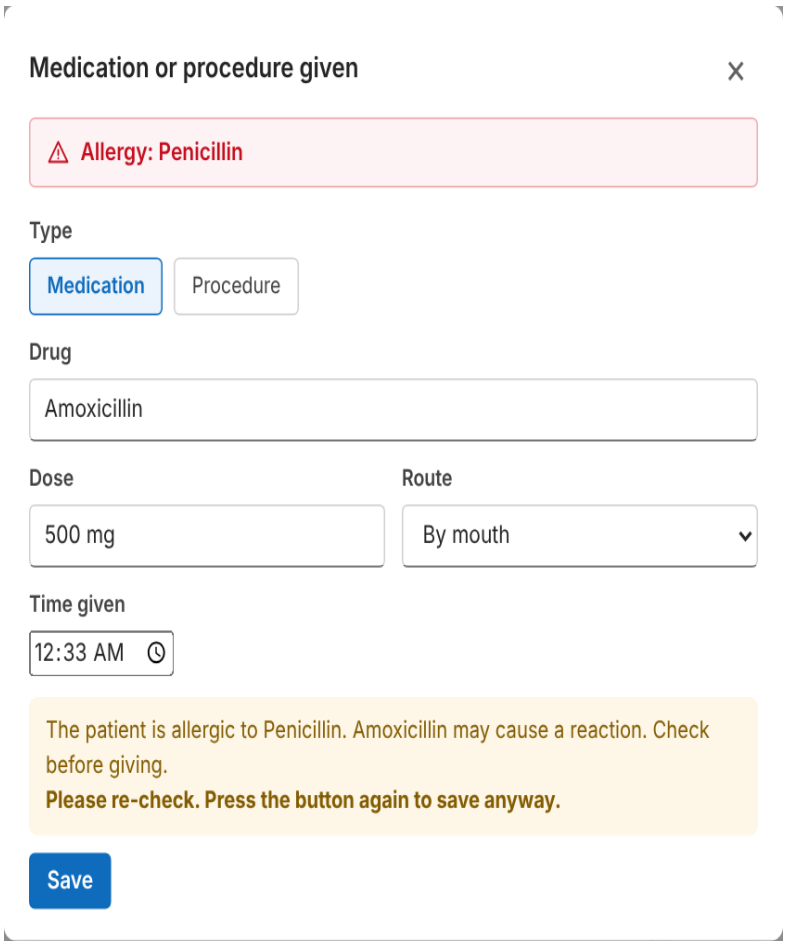



*Figure 18: Recording medication, with the allergy warning*

Choose Medication (drug, dose, route and time given) or Procedure (name and time). If the drug matches a known allergy, including penicillin-type antibiotics for a penicillin allergy, a warning appears before saving. If allergies are still unknown, the form asks to check first.

### 7.2.9 Reassess

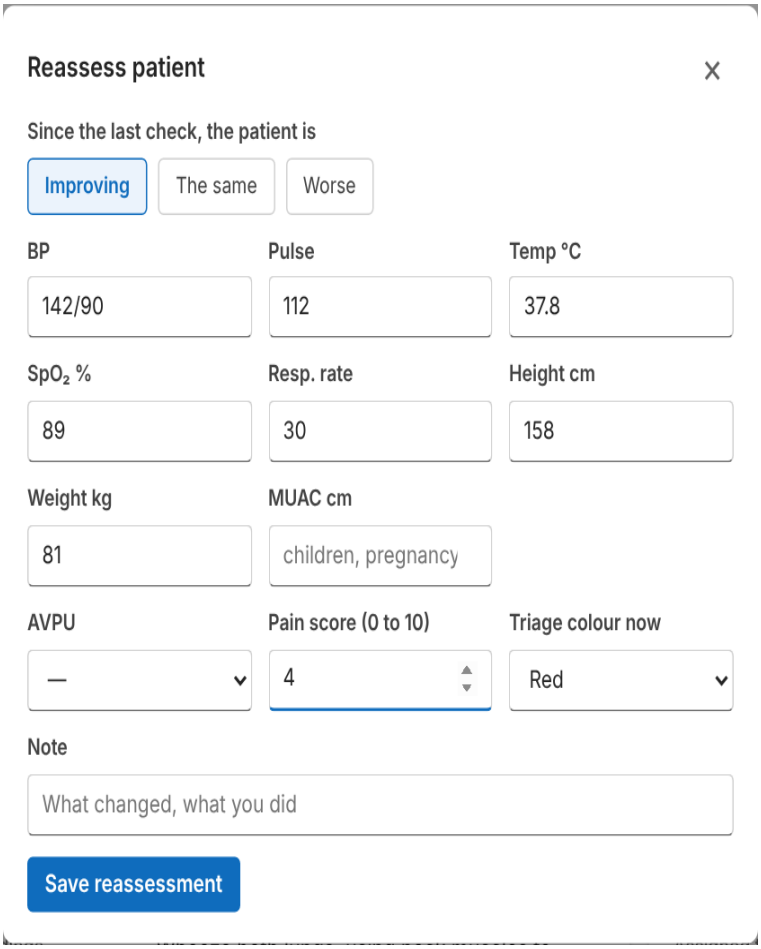



*Figure 19: Reassessing a patient*

Choose Improving, The same or Worse, and update the vital signs, AVPU, pain score (0 to 10) and the triage colour now. If the patient is worse but the colour is not raised, a warning suggests a higher colour. A change of colour is recorded in the journey, so triage can now be changed after registration.

### 7.2.10 Refer to another facility

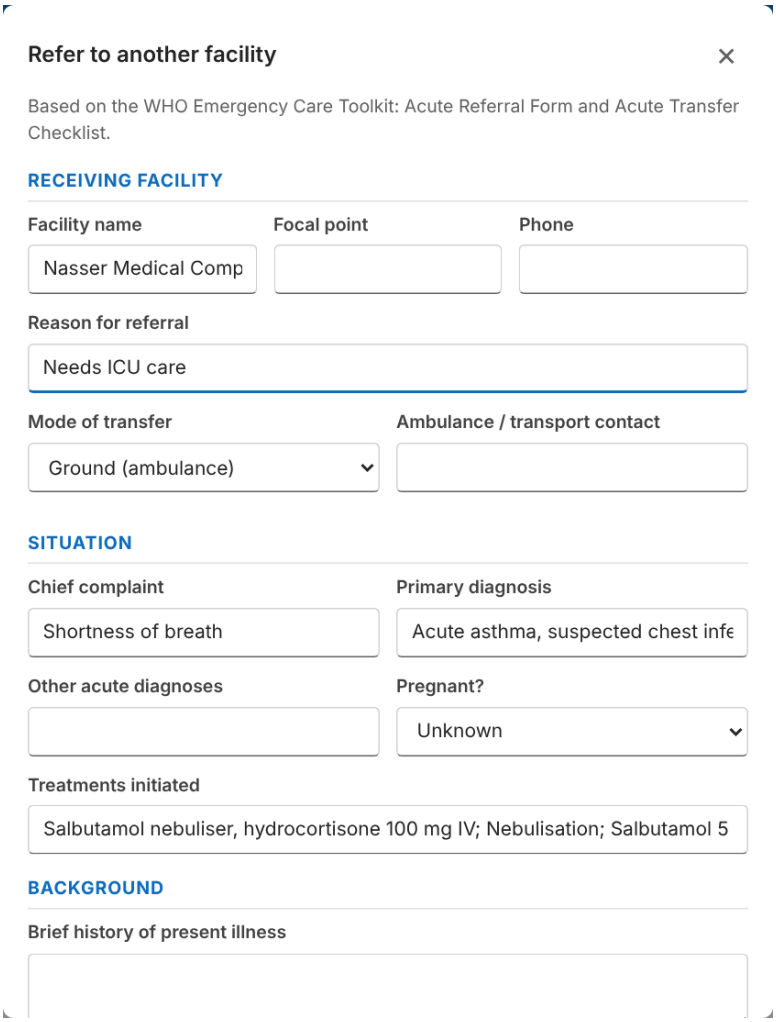
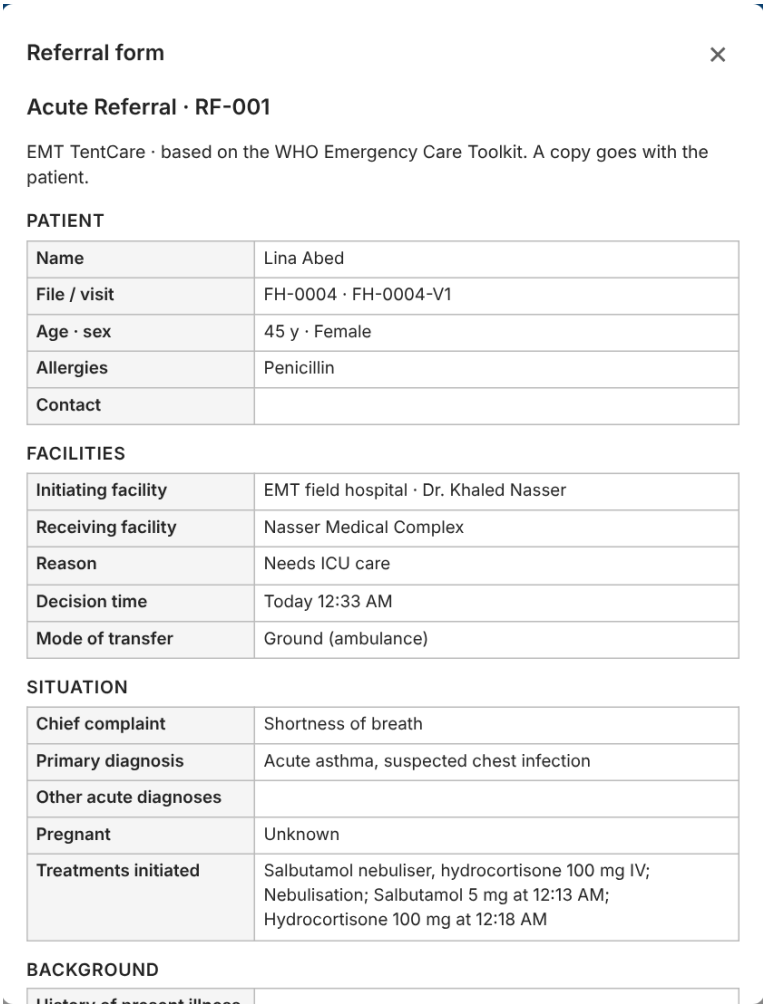



*Figure 20: The external referral form*

For a patient who needs care the field hospital cannot give. The form follows the WHO Acute Referral Form [2] and is filled from the file where possible:

- **Receiving facility:** name, focal point, phone, reason, mode of transfer (ambulance, other vehicle, air, sea) and transport contact.
- **SBAR:** Situation (complaint, diagnosis, pregnancy, treatments given), Background (history, ABCDE from the primary survey), Assessment (with current vital signs) and Recommendations (next steps, possible worsening, cautions).
- **Precautions:** infectious disease, spinal, weight-bearing, fall risk, aspiration risk, other.
- **Transfer checklist:** the 13 items of the WHO Acute Transfer Checklist (decision and preparation). If any item is not ticked, a warning says to contact a senior clinician; a second press saves.



Referral form

Acute Referral · RF-001

EMT TentCare · based on the WHO Emergency Care Toolkit. A copy goes with the patient.

PATIENT

| | |
|---|---|
| Name | Lina Abed |
| File / visit | FH-0004 · FH-0004-V1 |
| Age · sex | 45 y · Female |
| Allergies | Penicillin |
| Contact | |

FACILITIES

| | |
|---|---|
| Initiating facility | EMT field hospital · Dr. Khaled Nasser |
| Receiving facility | Nasser Medical Complex |
| Reason | Needs ICU care |
| Decision time | Today 12:33 AM |
| Mode of transfer | Ground (ambulance) |

SITUATION

| | |
|---|---|
| Chief complaint | Shortness of breath |
| Primary diagnosis | Acute asthma, suspected chest infection |
| Other acute diagnoses | |
| Pregnant | Unknown |
| Treatments initiated | Salbutamol nebuliser, hydrocortisone 100 mg IV; Nebulisation; Salbutamol 5 mg at 12:13 AM; Hydrocortisone 100 mg at 12:18 AM |

BACKGROUND

*Figure 21: The printable referral sheet*

After saving, the file shows the referral steps. Referral form prints the sheet that goes with the patient, with a part for the receiving facility. Then:

- **Patient has left:** records the departure. The outcome becomes Transferred to another facility and any bed goes to cleaning.
- **Record arrival:** the arrival time, who received the patient, and the status on arrival: Stable, Unstable or Dead.
- **Counter-referral received:** what the receiving facility sends back: care given (ICU, surgery, hospitalised), final diagnoses, next steps, pending tests, follow-up timeframe and what to do if the patient gets worse.

### 7.2.11 Update allergies and Start new visit

Update allergies changes the allergy status at any time; the old and new values are recorded. Start new visit opens a new visit for a discharged patient who comes back, on the same file number.

## 7.3 Patient journey

Every action is recorded automatically in the patient’s journey with the time and the staff member’s name: registration and triage criteria, referral and acceptance, start of consultation, primary survey, assessment, medication and procedures, vital signs and reassessment, test request and result, admission and arrival on the ward, external referral steps, discharge approval and departure, and

follow-up attended or missed. Staff cannot edit it by hand. It is the basis of the audit trail in the real version.

# 8. Lab and X-ray

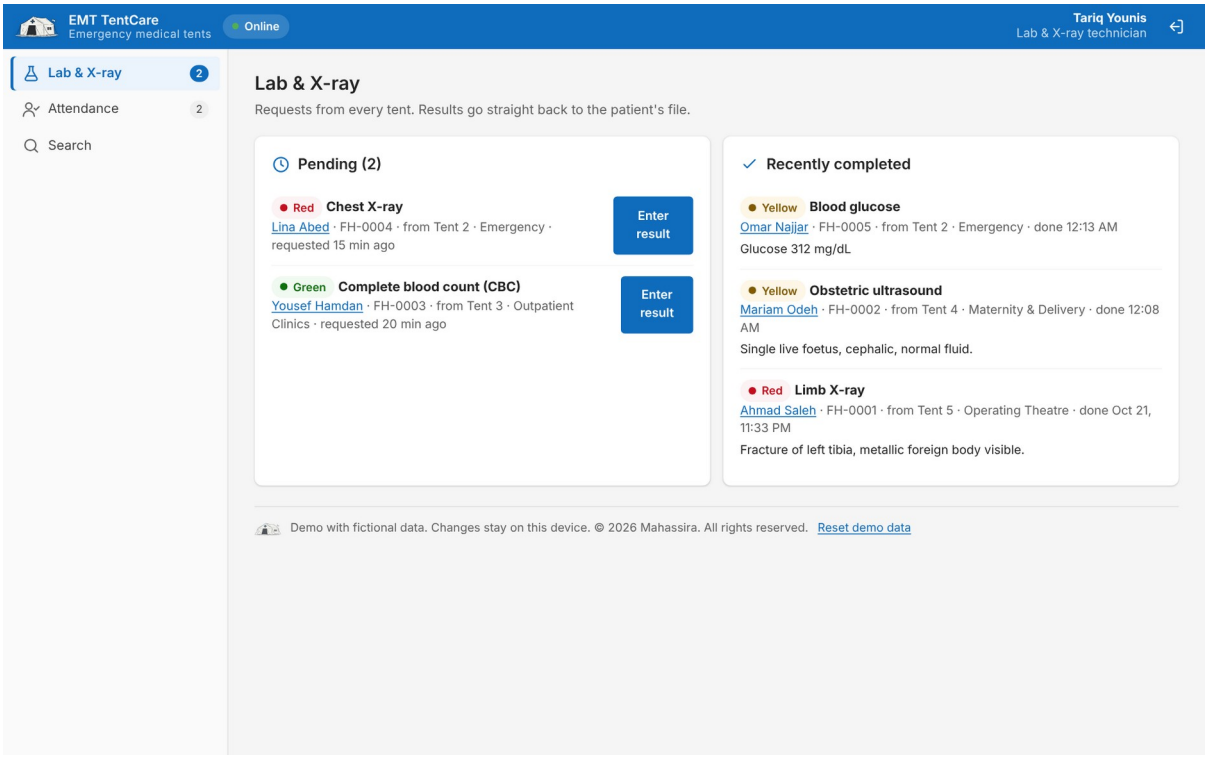



*Figure 22: The Lab & X-ray screen*

The **Lab & X-ray** screen has two parts:

- **Pending:** every request still waiting for a result, from every tent. Sorted by the patient's triage colour first, then the oldest request. Each request shows the test, the patient's name and file number, the requesting tent, and how long ago it was requested.
- **Recently completed:** the last 10 results, with the result and its time.

To enter a result: press **Enter result** next to the request, type the result, then press **Send result to the patient's file**.

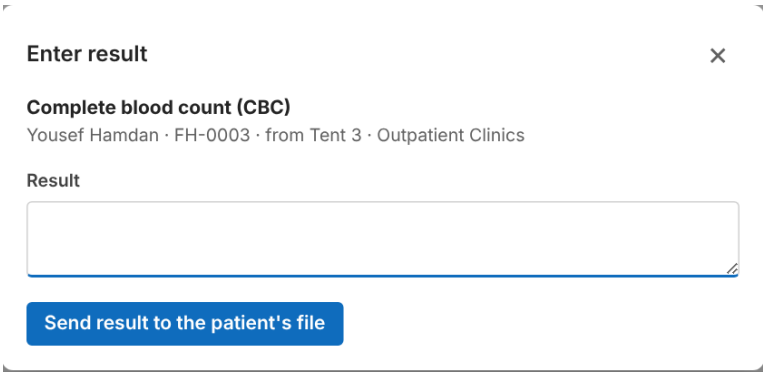



*Figure 23: Entering a result*

The result shows at once in the Lab & imaging section of the patient file, and a "Results ready" badge appears on the patient's card in the tent queue.

# 9. Inpatient ward and beds

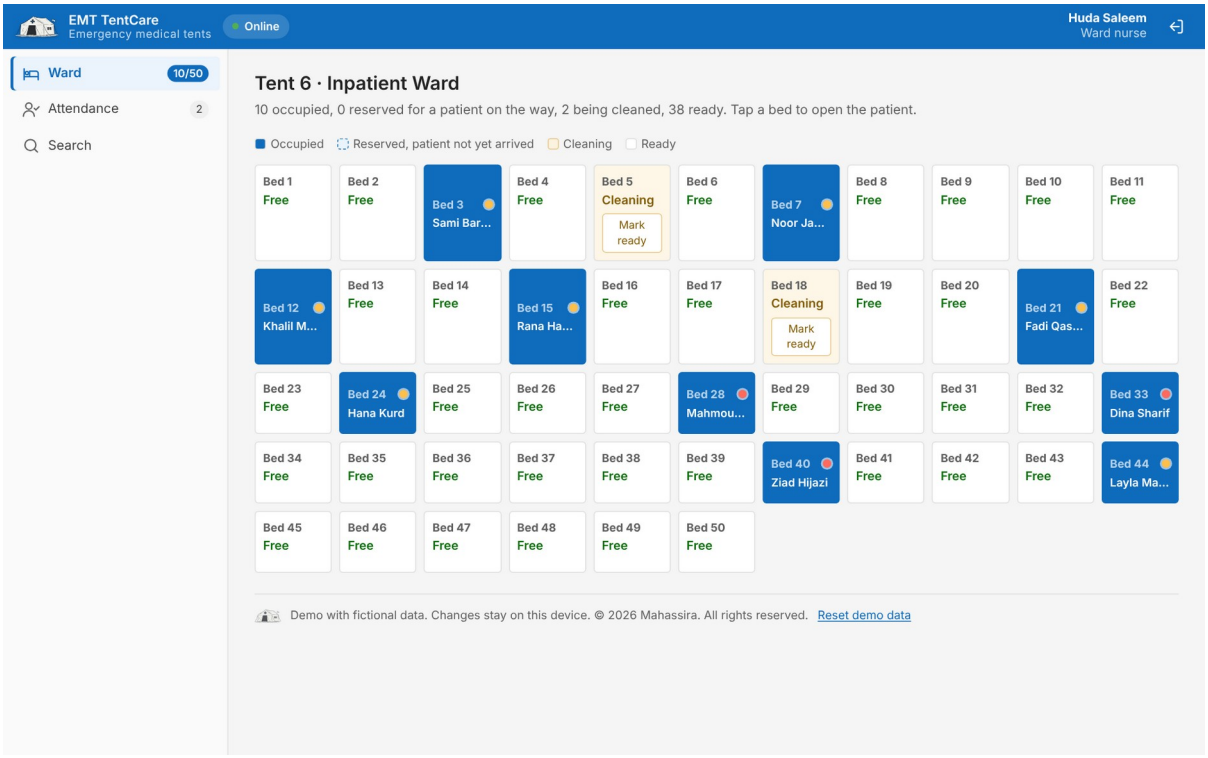



*Figure 24: The map of the 50 beds*

- The screen shows the 50 beds as numbered squares.
- **An occupied bed** is blue, with the patient's name and a dot in their triage colour. Tapping it opens the patient's file.
- A **reserved** bed has a dashed blue border and the patient's name: the patient is admitted but not yet on the ward.
- A bed **being cleaned** is light orange and marked Cleaning. The nurse or site manager presses Mark ready when it is clean.
- A **ready** bed is white and marked Free. Only ready beds can be chosen for an admission.
- The top of the screen shows how many beds are occupied, reserved, being cleaned and ready, with a colour key.
- When a patient leaves the ward (discharge, transfer or referral), the bed goes to Cleaning automatically.

# 10. Attendance for staff and patients

The Attendance screen is available to every role. It has two tabs: **Staff** and **Patients**.

## 10.1 Staff attendance

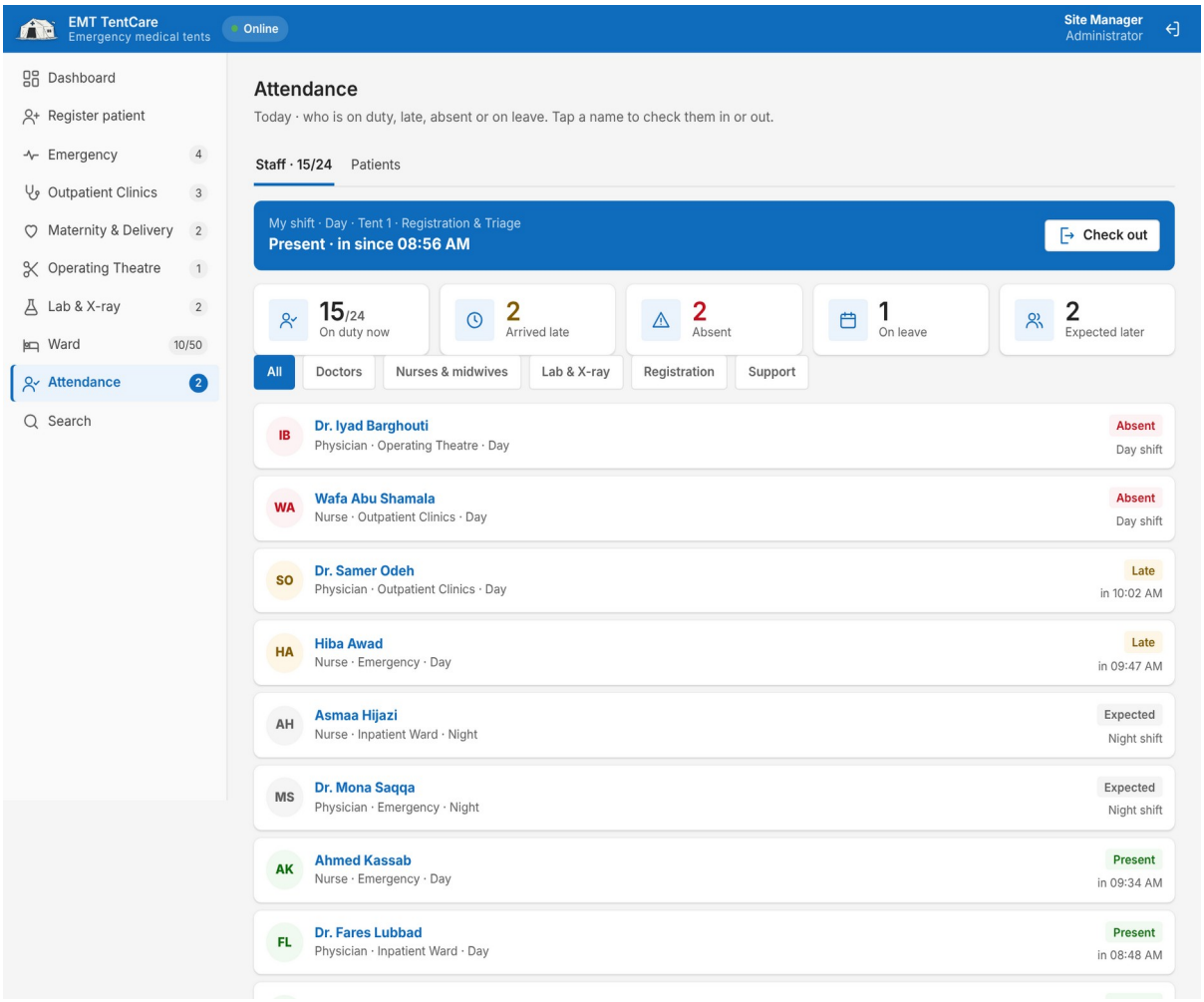



*Figure 25: Staff attendance*

The demo data has 24 staff: doctors, nurses, midwives, lab technicians, registration clerks and support staff (cleaners, a guard, a driver and the site manager). Each has a tent and a Day or Night shift.

### Attendance statuses

*Table 16: Staff attendance statuses*

| Status | Meaning | Colour |
|---|---|---|
| Present | On shift. | Green |
| Late | On shift but arrived late. | Orange |
| Absent | Absent. | Red |
| On leave | On leave. | Purple |
| Checked out | Finished the shift and left. | Blue |

| Status | Meaning | Colour |
|---|---|---|
| Expected | Not arrived yet: the shift hasn't started or the person is still expected. | Grey |

**When someone counts as late**

- **Day shift:** checking in after 8:15 am.
- **Night shift:** checking in after 8:15 pm.

**What the screen shows**

- **Blue "My shift" card:** the current user's shift and status, with a Check in or Check out button.
- **Tiles:** on duty out of the total, late arrivals, absent, on leave, and expected later.
- **Filter chips:** All, Doctors, Nurses & midwives, Lab & X-ray, Registration, Support.
- **Staff list:** absent and late staff first. Each row shows the name, job, tent, shift, status and check-in or check-out time.

**Changing someone's attendance**

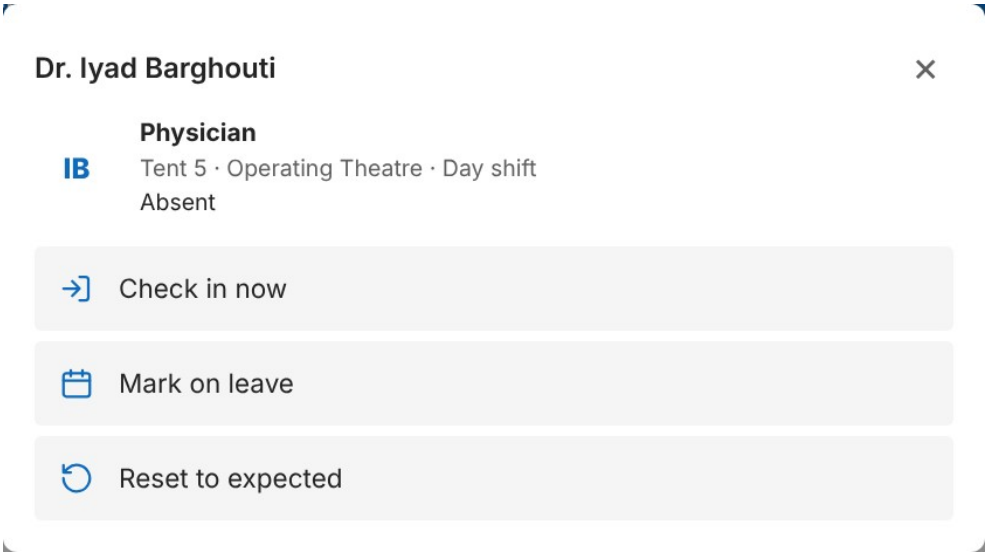



*Figure 26: A staff member's attendance options*

Tapping a name opens a panel with Check in now or Check out now, Mark absent, Mark on leave, and Reset to expected. **Every staff member can check themselves in and out.** Changing other people's attendance, and recording absence and leave, is for the registration desk and the site manager. Anyone can also check in from the account menu.

The attendance record starts fresh every day automatically.

## 10.2 Patient attendance and follow-up

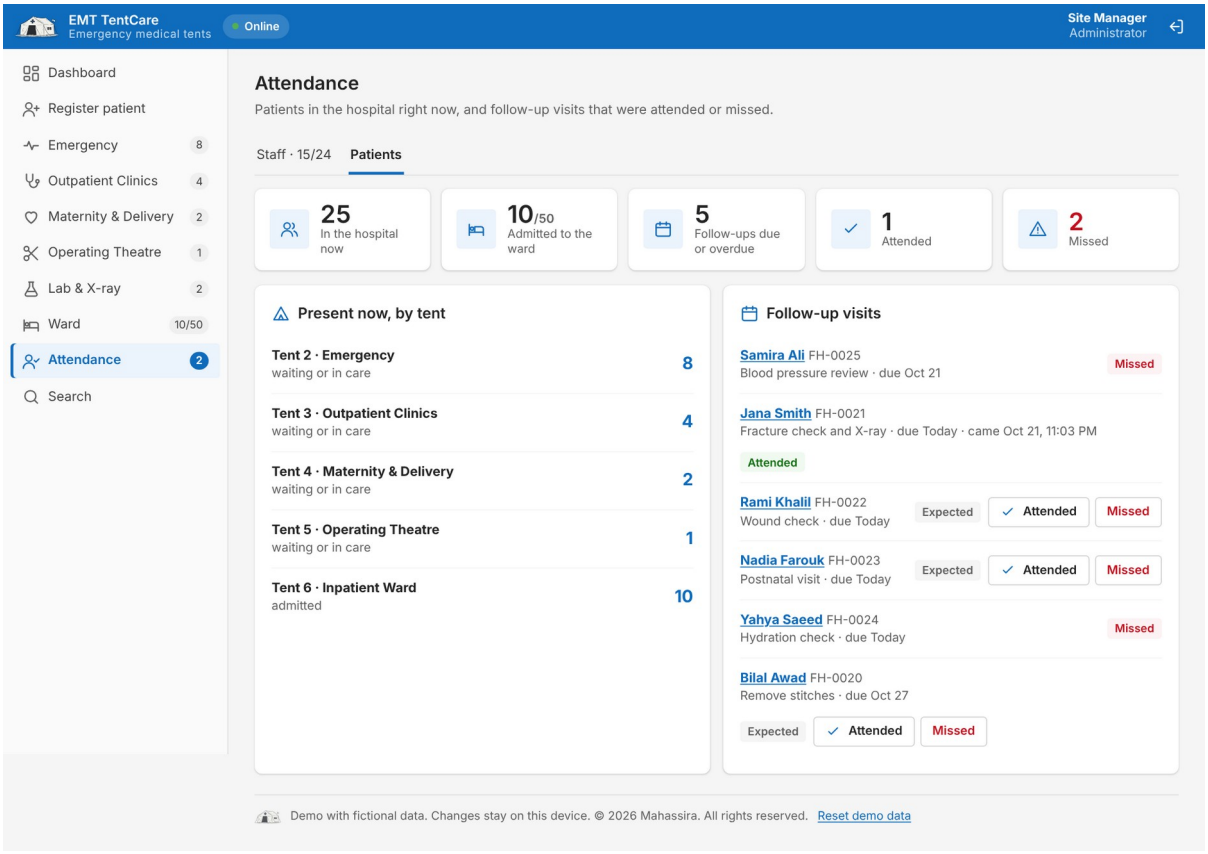



*Figure 27: Patient attendance and follow-up visits*

- **Tiles:** patients in the hospital now, patients admitted to the ward, follow-ups due or overdue, attended, and missed.
- **Present now, by tent:** how many patients are in each tent right now. Tapping a tent opens its queue.
- **Follow-up visits:** every follow-up, sorted by date, with its reason and status: Expected, Attended or Missed.

When the patient comes, press **Attended**. If they don't, press **Missed**. **A follow-up whose date has passed without being marked attended shows automatically as Missed.** Each mark is added to the patient's record. Every role except the lab can mark follow-ups.

# 11. Site manager dashboard

The **Dashboard** is the site manager's first screen. It gives a complete picture of the hospital from the device's data.

*Table 17: Dashboard tiles*

| Tile | Meaning |
|---|---|
| Patients in the hospital | Everyone not yet discharged. |
| Red (immediate) cases | Red cases in the hospital now. |
| Mass casualty patients | Shows only during a mass casualty incident. |
| Ward beds occupied | Occupied beds out of 50. |
| Lab & X-ray results pending | Requests waiting for a result. |
| Registered today | Patients registered today. |
| Discharged or transferred today | Patients who left or were transferred today. |
| Staff on duty | Staff present now, out of the total. |
| Follow-ups due today | Follow-up visits due today. |

- **Patients by tent:** a bar for each tent showing how many are waiting and how many are in care. Tapping a tent opens it.
- **Latest activity:** the latest actions across the hospital with time and staff member. Tapping a patient's name opens their file.
- **Export to Excel:** exports all the data to an Excel file (chapter 13).

# 12. Finding a patient

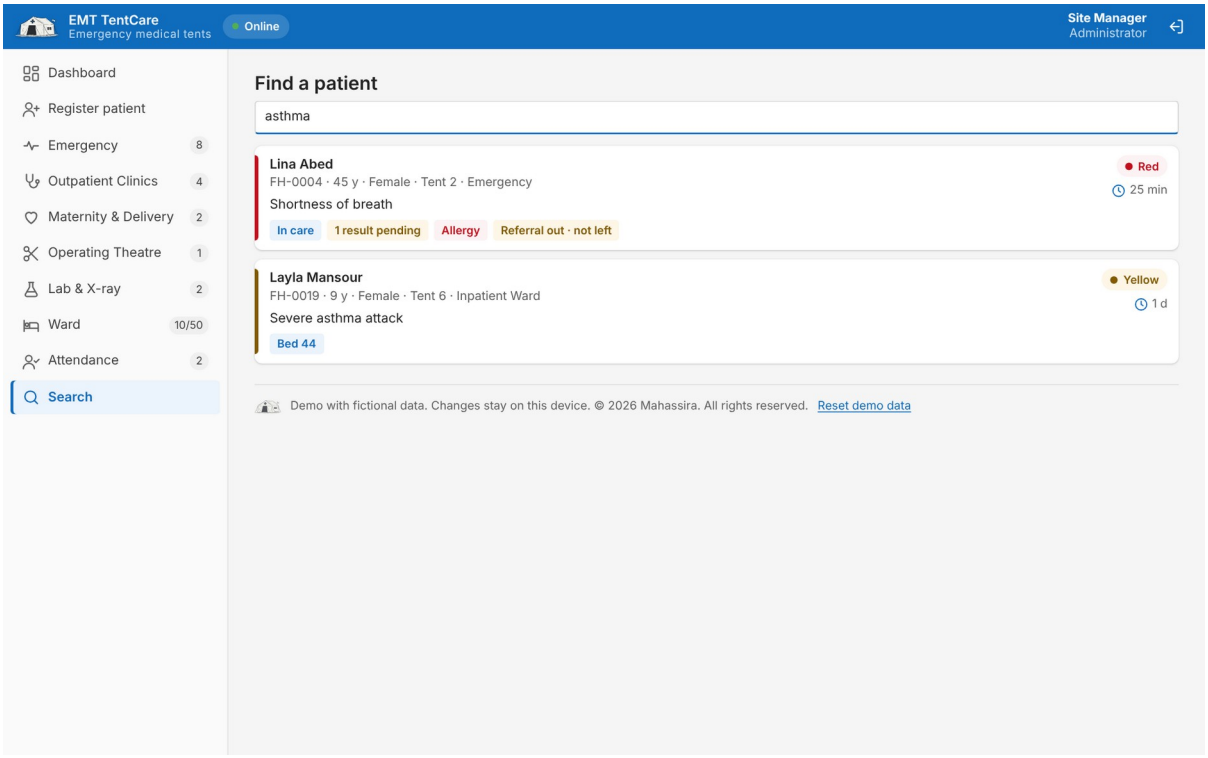



*Figure 28: Searching by diagnosis*

The **Search** screen is available to every role. Search by name, file number or diagnosis. Results appear as you type: up to 30, newest first, with where each patient is now. Discharged patients are included. On a phone, search opens from the magnifier in the top bar.

# 13. Exporting data to Excel

On the Dashboard, **Export to Excel** creates a file named TentCare-date.xlsx. The file is built on the device itself, so export works without internet.

*Table 18: Sheets in the Excel export*

| Sheet | Contents |
|---|---|
| Patients | One row per patient: personal details, triage and triage criteria, mass casualty temporary ID, vital signs, clinical assessment, visit outcome, referral, follow-up, assigned doctor and last update. |
| Previous visits | Earlier visits of returning patients. |
| Primary surveys | Every ABCDE survey: one column per letter, GCS, mechanism and intent. |
| Medication & procedures | Every drug and procedure given, with dose, route, time and who gave it. |
| Reassessments | Every reassessment with vital signs, AVPU, pain and triage before and after. |
| External referrals | Every referral to another facility: facility, reason, mode, checklist, times, status on arrival and the counter-referral. |
| Lab & X-ray | Every request: test, patient, tent, time, status and result. |
| Ward beds | The 50 beds: occupied, reserved, cleaning or ready, the patient and their diagnosis. |
| Staff attendance | Today’s staff attendance: status, lateness, check-in and check-out times. |
| Follow-ups | Follow-up visits and their status. |
| Activity log | The full log of every action, in time order. |

On every sheet the header row is blue and stays in place when scrolling, with filters ready to use.

# 14. Working online and offline

The system is designed to work **online and offline**, because the internet in the field cannot be relied on.

## 14.1 How the demo works

- The first time the link is opened with internet, the browser saves a copy of the system on the device using a service worker.
- After that the system opens and works fully without internet.
- All data is saved in the browser's storage on the device itself.
- The connection indicator in the top bar switches from Online to Offline when the internet drops.
- The system can be installed like an app: on iPhone, from Safari's share button then Add to Home Screen; on Android, from the Chrome menu then Install app or Add to Home screen.

> **Limits of the demo:** data stays on the device and browser where it was entered and does not move to other devices. Clearing the browser's data deletes it. The demo is for showing and training only.

## 14.2 How the real version will work

1. **With a connection:** every registration, result and discharge is saved to the central database (Microsoft Dataverse) straight away, and every staff member sees it on every device.
2. **When the internet drops:** staff keep working on the same phone or tablet. Each change is saved on the device and queued.
3. **When the connection returns:** the device sends all changes automatically to the central database and receives other people's changes, with nothing to re-enter.
4. **Devices:** managed with Microsoft Intune, so they can be locked with a PIN and wiped remotely if lost.

Technically this is possible in Power Apps with its offline support for Dataverse [6]. How data stored on the device is protected while offline is one of the points to test in the first pilot with the IT team.

# 15. Data security and protection

## 15.1 Where things stand, honestly

*Table 19: Security: demo version and proposed real version*

| Area | Demo version | Proposed real version |
|---|---|---|
| Sign-in | Demo accounts with one password shown on screen. | Each staff member's work account through Microsoft Entra ID, with multi-factor authentication (MFA). |
| Where data lives | In the device's browser only, unencrypted. | Microsoft Dataverse inside the organisation's environment, encrypted at rest and in transit. |
| Permissions | Five roles built into the page. | Security roles in Dataverse, managed by the IT team. |
| Audit trail | Exists, but can be deleted by clearing browser data. | Dataverse auditing linked to the staff member's email; cannot be changed from the device. |
| Export | Unprotected Excel file. | Power BI reports with aggregated numbers, or files protected by the organisation's policies. |
| Data sent outside | Sends no data to any server or outside service. | Data does not leave the organisation's Microsoft 365 environment. |

## 15.2 Security layers in the real version

*Table 20: Security layers in the real version*

| Layer | What it protects | Tool |
|---|---|---|
| Identity | Only the organisation's staff can sign in; accounts close when staff leave. | Microsoft Entra ID, MFA, Conditional Access |
| Permissions | Each role sees only what it needs; each site is separated. | Dataverse security roles and business units |
| Sensitive fields | Fields such as GBV care visible only to a named team. | Column-level security |
| Encryption | Data encrypted at rest and in transit. | Microsoft-managed keys, or a key the organisation owns in Azure Key Vault |
| Devices | A lost or seized device can be wiped remotely. | Microsoft Intune |
| Audit | Who opened or changed which file, and when. | Dataverse auditing |
| Leak prevention | Data cannot be sent to unapproved services. | Power Platform DLP policies |
| Backup | Data can be restored after a mistake. | Automatic Dataverse environment backups |

## 15.3 Humanitarian data protection principles

- **Do no harm:** patient data in a conflict zone can put people at risk if it leaks. Reports to donors and the Health Cluster are aggregated numbers with no names.
- **Data minimisation:** phone and shelter are optional, and the organisation decides which fields are required.
- **Consent:** a consent field is added at registration in the real version.
- **Retention:** when the response ends, data is archived or deleted according to the organisation's policy.
- **Ownership:** the data belongs entirely to the organisation.

Reference: IASC Operational Guidance on Data Responsibility in Humanitarian Action (2023) [4] and the ICRC Handbook on Data Protection in Humanitarian Action [15].

# 16. Technical design and the move to Microsoft 365

## 16.1 Tools used for the demo

*Table 21: Tools used for the demo*

| Tool | Role |
|---|---|
| HTML, CSS and JavaScript | The whole system is one file, with no outside libraries or services. |
| Service worker | Saves the system on the device so it works offline, and lets it be installed as an app (PWA). |
| localStorage | Stores data in the device's browser. |
| Excel export | Built into the system itself; works offline. |
| GitHub and GitHub Pages | Keep the code and its history, and host the demo link. |
| Playwright | Automated test of the full patient journey on phone, tablet and computer before every update. |
| Claude by Anthropic | AI tool used to write the code (AI-assisted development), based on the project owner's requirements and field experience. |
| Canva | Logo design. |

The current design follows Microsoft's Fluent style, with the colours, font and buttons of Microsoft 365 apps, so it is close to how the real Power Apps version will look.

## 16.2 What each part becomes in Power Platform

*Table 22: From the demo to Power Platform*

| In the demo | In the real version |
|---|---|
| Screens and interface | Power Apps canvas app with the same colours and layout |
| Data in the browser | Microsoft Dataverse tables |
| Demo accounts | Sign-in with work accounts (Microsoft Entra ID) |
| The five roles | Dataverse security roles |
| Patient journey | Dataverse auditing and a log table |
| Offline use | Power Apps offline with Dataverse |
| Dashboard and Excel export | Power BI and Excel export |
| Alerts (bed needed, red case) | Power Automate with Microsoft Teams and Outlook |
| Documents and X-ray images | SharePoint |

## 16.3 Proposed Dataverse tables

*Table 23: Proposed Dataverse tables*

| Table | Contents |
|---|---|
| Patients | Core patient details, status, tent and bed. |

| Table | Contents |
|---|---|
| Clinical Assessments | Clinical assessments. |
| Vital Signs | Vital sign readings with time. |
| Lab & Imaging Orders | Requests and results. |
| Beds | The 50 beds and their status. |
| Staff | Staff, job, tent and shift. |
| Attendance | Daily attendance records. |
| Follow-ups | Follow-up visits. |
| Activity Log | Log of actions. |

Plus fixed choice lists (Choices): tents, triage colours, body systems, emergency categories, procedures, visit outcomes and referrals.

# 17. Demo data

The first time the demo opens it fills itself with fictional data so it looks like a working hospital:

- About 25 patients across all tents, with different statuses and triage colours.
- 10 patients in the ward on different beds.
- Test and imaging requests, some with results and some pending.
- Discharged patients and follow-up visits: some attended, some missed, some still expected.
- 24 staff on day and night shifts, some present, some late, some absent and some on leave.
- Lina Abed (FH-0004) has an example primary survey and medication, and two ward beds are waiting to be cleaned.

All changes stay on the device. To return to the starting data, press **Reset demo data** at the bottom of the page on a computer, or from the account menu on a phone. Useful before any demo or training session.

# 18. Quick guide for each role

Each section can be printed and put up in the tent.

**Registration clerk**

1. Sign in and press Check in from the account menu or the Attendance screen.
2. Open Register patient.
3. Type the name and age, pick the gender, write the complaint, and choose the allergy status. If a “Possible existing patient” window appears, check with the patient.
4. Take the vital signs and enter them.
5. Open Triage criteria and tick any sign that applies, then pick the colour. Follow the suggestion unless there is a clear reason not to.
6. Pick the tent and press Register and send.
7. Print the wristband or write the file number on it, then press Done.
8. To answer questions about a patient: Today’s patients or Search.
9. Many patients at once: press Mass casualty mode and use fast registration.

**Physician**

1. Open your tent’s queue. The patient at the top is the one to see first.
2. Tap the patient, then Start consultation.
3. Read the red allergy bar if it shows.
4. For emergencies, record the Primary survey (ABCDE). Then fill in Clinical assessment, and record each medication or procedure as it is given.
5. If needed: Request lab / X-ray, and watch for the Results ready badge.
6. Decide: Refer to another tent, Admit to ward, Refer to another facility, or Approve discharge. When a patient is sent to your tent, press Accept patient.
7. On discharge: pick the visit outcome and write the take-home medication, referral and follow-up. Outside the ward, confirm the departure when the patient leaves.

**Lab & X-ray technician**

1. Open Lab & X-ray.
2. Start with the top request in Pending: it is the most serious and the oldest.
3. Press Enter result, type the result, then Send result to the patient’s file.

**Ward nurse**

1. Open Ward to see the beds.
2. Tap a bed to open the patient’s file.
3. When a new patient comes, press Patient arrived on ward. On every round use Reassess or Update vitals, and record any medication given.
4. When the doctor has approved a discharge and the patient leaves, press Confirm departure. Clean the bed, then press Mark ready on the ward map.

**Site manager**

1. Start the day on the Dashboard: red cases, beds and pending results.
2. Check Attendance: who is absent or late, and record leave.
3. Check follow-up visits in the Patients tab.
4. At the end of the day: Export to Excel for the report.

# 19. Frequently asked questions

**What happens if the internet drops while I’m working?**

Nothing stops. The system keeps working and saves everything on the device, and the connection indicator changes to Offline.

**Can a colleague on another device see what I entered?**

In the demo, no: data stays on each device. In the real version, yes, as soon as there is a connection.

**How do I know a test result is back?**

A Results ready badge appears on the patient’s card in the queue, and the result is in the Lab & imaging section of the file.

**What if I picked the wrong triage colour?**

The doctor or ward nurse presses Reassess and picks the new colour. The change is recorded in the patient’s journey.

**Can a discharge be undone?**

An approved discharge can be cancelled by the doctor until the patient leaves. Once the departure is confirmed, the file is read-only. If the patient comes back, registration opens a new visit on the same file.

**What happens when the ward is full?**

A “Ward is full” message appears when trying to admit. Beds being cleaned do not count as free. A patient must be discharged or transferred, or a bed marked ready, first.

**Can the demo be used with real patients?**

No. It has no real sign-in and no encryption. It is for showing and training only.

**Who owns the data?**

In the real version, the organisation owns all of it, inside its own Microsoft 365 environment.

**Are the clinical forms official WHO forms?**

No. They are based on the WHO Emergency Care Toolkit, which is free to use, but the system is not an official WHO tool and is not endorsed by WHO. The organisation’s clinical leads should review them before real use.

**What if a mass casualty patient’s name is never known?**

The patient keeps the temporary ID (for example MCI-004) and the normal file number. Everything recorded stays linked to that file.

# 20. Glossary

*Table 24: Glossary of terms*

| Term | Meaning |
|---|---|
| ABCDE / primary survey | A quick check in order: Airway, Breathing, Circulation, Disability, Exposure. |
| Admit | Admit the patient to the inpatient ward. |
| Audit trail | A record of who did what, and when. |
| AVPU | Level of consciousness: Alert, responds to Voice, responds to Pain, Unresponsive. |
| Counter-referral | The reply a receiving facility sends back about a referred patient. |
| Dataverse | Microsoft's database inside Power Platform. |
| Dead on arrival | The patient died before reaching the hospital. |
| Discharge | The patient leaves the hospital. |
| EMT (Emergency Medical Team) | A medical team deployed in disasters and conflicts. |
| EMT Type 1, 2, 3 | WHO levels for Emergency Medical Teams: Type 1 outpatient care, Type 2 inpatient surgical care, Type 3 referral care with intensive care. |
| FAST | A quick ultrasound to look for blood in the abdomen and chest after injury. |
| File number | The patient's file number, such as FH-0026. |
| Follow-up | A visit after discharge. |
| GBV | Gender-based violence. |
| GCS | Glasgow Coma Scale, 3 to 15: eyes (E), voice (V) and movement (M). |
| Health Cluster | The coordination group for health organisations in a humanitarian response. |
| Intune | Microsoft's tool for managing and protecting devices. |
| Mass casualty incident (MCI) | More patients at once than the hospital can handle normally. |
| Microsoft 365 | Microsoft's apps for organisations: Outlook, Teams, SharePoint, Excel and more. |
| Microsoft Entra ID | The identity and work-account sign-in service. |
| MUAC | Mid-upper arm circumference, used to screen for malnutrition. |
| Offline-first | A system built to work without internet first, then sync when connected. |
| Power Apps | Microsoft's tool for building apps. |
| Power Automate | Microsoft's tool for automating tasks and alerts. |
| Power BI | Microsoft's tool for reports and dashboards. |
| PWA | A web app that can be installed on a device and works offline. |
| Referral | Sending the patient to another tent or service. |

| Term | Meaning |
|---|---|
| SBAR | Situation, Background, Assessment, Recommendation: a standard way to hand over a patient. |
| $SpO_2$ | Oxygen saturation of the blood. |
| SRH | Sexual and reproductive health. |
| Triage | Sorting patients by how serious their condition is. |
| Vitals | Vital signs: blood pressure, pulse, breathing, oxygen and temperature. |
| WHO Emergency Care Toolkit | Free forms and tools from the World Health Organization for emergency units. |

# 21. Current limits and next steps

## 21.1 Limits of the current version

- Data lives on one device and is not shared between devices.
- No real sign-in and no encryption.
- Registration details such as the name cannot be edited after saving (except identifying a mass casualty patient).
- No automatic alerts to staff.
- The interface is in English only. The WHO-based forms and triage rules have not yet had a clinical review by the organisation.
- No field for the patient's consent to data collection.

## 21.2 Proposed next steps

1. **Review:** the organisation's IT team and data protection lead review the design and decide which fields are allowed.
2. **Test environment:** a separate Power Platform environment inside the organisation's Microsoft 365.
3. **Build:** the real version in Power Apps and Dataverse with fictional data, including roles and the audit trail.
4. **Field pilot:** 5 to 10 staff try it on Intune-managed devices, including offline use.
5. **Decision:** after the pilot, management decides whether to use it with real patients.

## 21.3 Proposed improvements

- Interface in both Arabic and English.
- A clinical review of the WHO-based forms and triage rules by the organisation's doctors.
- Teams alerts when a red case arrives or a bed is needed.
- Daily and weekly Power BI reports with aggregated numbers.
- A patient consent field, and rules to hide sensitive fields.

# 22. IT, security and monitoring & evaluation

This chapter is for the IT team, the data protection lead and the monitoring and evaluation (M&E) team. It explains how the real version would be built entirely on Microsoft 365 and Power Platform, how patient data and files are protected, and how the system supports reporting. Everything here stays inside the organisation's own Microsoft tenant.

## 22.1 How the system is built on Microsoft

*Table 25: Microsoft services used in the real version*

| Part | Microsoft service | What it does |
|---|---|---|
| App | Power Apps (canvas app) | The screens staff use on phones, tablets and computers. Works offline on the Power Apps mobile app. |
| Database | Microsoft Dataverse | Stores patients, visits, orders, beds, messages and the activity log. |
| Sign-in | Microsoft Entra ID | Work accounts, two-step sign-in and access rules. |
| Automation | Power Automate | Email alerts, escalation of unanswered calls, daily reports. |
| Files | SharePoint (or Dataverse file columns) | X-ray images, referral letters and scanned documents. |
| Reports | Power BI | Dashboards for managers and M&E, with totals only. |
| Notifications | Outlook and Teams | Alerts to staff, with no patient details. |
| Devices | Microsoft Intune | Manages phones and tablets; lock and wipe if lost. |
| Compliance | Microsoft Purview | Sensitivity labels, audit search and retention rules. |
| Administration | Power Platform admin center | Environments, data loss prevention policies, capacity and backups. |

## 22.2 Environments and changes

The system is never built in the organisation's default environment. IT creates three separate environments, each with its own Dataverse database:

- **Development:** where the app is built and changed. Only made-up data.
- **Test:** where staff try each new version before release. Only made-up data.
- **Production:** the live system with real patients. Nobody edits the app here directly.

The app is packaged as a Power Platform solution and moved from Development to Test to Production. Every release is approved, recorded and can be rolled back. The production environment should be a Managed Environment, which adds extra controls such as sharing limits, usage insights and longer backups.

## 22.3 Who can sign in, and what they can see

- Every staff member signs in with their own work account. There are no shared accounts.
- **Two-step sign-in (MFA)** is required for everyone.

- **Conditional Access** rules allow sign-in only from managed devices and block risky sign-ins [16].
- Each role (registration, doctor, nurse, lab, site manager, M&E) is a **security role** in Dataverse, given through an Entra ID group. IT manages the groups, not the app.
- Each field site is a separate **business unit**, so staff only see patients from their own site.
- Sensitive fields, such as GBV care or HIV status, use **column-level security**, so only a named clinical team can see them.
- When someone leaves, their account is disabled the same day and they lose access on every device.
- Access is reviewed every month by the site manager and IT.

## 22.4 Protecting data and files

*Table 26: How data and files are protected*

| Where | Protection |
|---|---|
| In the database | Dataverse encrypts all data at rest. By default Microsoft manages the key; the organisation can use its own key stored in Azure Key Vault [7]. |
| On the network | All traffic between devices and Microsoft is encrypted (TLS). |
| Files and images | Stored in a SharePoint library or Dataverse file column that only the clinical team can open. Purview sensitivity labels (for example “Confidential – Patient data”) encrypt each file, and the protection stays with the file even if it is downloaded. |
| Exports | Excel exports carry the same sensitivity label. Download and printing can be blocked on devices that are not managed. |
| Email and Teams | Notifications never contain patient names or details, only the file number and a link to open the app. |
| On the device | Offline data stays inside the Power Apps mobile app. Intune app protection stops copying patient data into personal apps or storage. |

## 22.5 Devices and offline use

- All phones and tablets are enrolled in Intune with a PIN or fingerprint and device encryption.
- A lost or seized device can be locked and wiped remotely.
- App protection policies block copy and paste, screenshots and saving files outside managed apps.
- How offline data is protected on the device, and how sync conflicts are handled, are tested in the first pilot before any real use.

## 22.6 Stopping data leaks

- **Data loss prevention (DLP) policies** in Power Platform allow only approved connectors (Dataverse, SharePoint, Outlook, Teams) and block social media and personal storage connectors [17].
- The app can only be shared with the staff security groups, not with guests.
- Tenant isolation stops data from being moved to other Microsoft tenants.

## 22.7 Monitoring by IT

- **Dataverse auditing** records who created, changed or viewed a patient record, and when [18]. Staff cannot change the audit log.
- The **Microsoft Purview audit log** shows sign-ins, file access and admin changes across Microsoft 365.
- Power Platform admin center shows app usage, errors and failed flows.
- Alerts go to IT for unusual activity, such as many records exported at once or sign-ins from unusual places.
- IT checks every week that offline devices have synced and that email alert flows are running.

## 22.8 Backup, recovery and continuity

- Dataverse takes automatic system backups. They are kept for 7 days by default and up to 28 days in Managed Environments [8].
- IT takes a manual backup before every release.
- If the whole system is down, staff use printed paper forms with the same fields and enter them later.
- A restore test is done at least once before go-live.

## 22.9 Keeping and deleting data

The organisation decides how long patient data is kept. When the response ends, data is archived or deleted according to that policy, using Purview retention rules. Every deletion is recorded. Data never leaves the organisation’s tenant.

## 22.10 Monitoring and evaluation (M&E)

Because every step is recorded with a time and a name, the M&E team can measure how the hospital works without extra data collection. Power BI dashboards update automatically from Dataverse.

*Table 27: Main M&E indicators*

| Indicator | How it is measured | Source in the system |
|---|---|---|
| Patients seen | New patients and visits per day, by tent, triage colour, age group and sex. | Registration |
| Waiting time | Median minutes from registration to start of consultation, by triage colour. | Activity log |
| Length of stay | Hours from arrival to departure; days on the ward. | Visits |
| Bed occupancy | Occupied beds divided by 50, each day. | Ward beds |
| Lab turnaround | Minutes from request to result. | Lab & X-ray |
| Referrals out | Number referred to other hospitals, and status on arrival. | External referrals |
| Outcomes | Discharged, transferred, left early, died, dead on arrival. | Visits |
| Follow-up | Share of follow-up visits attended. | Follow-ups |

| Indicator | How it is measured | Source in the system |
|---|---|---|
| Response to calls | Minutes until a call or critical result is acknowledged. | Messages |
| Data quality | Share of files with allergy status known and vital signs recorded. | Patients |

- Daily and weekly reports for donors and the Health Cluster use totals only. Numbers smaller than 5 are hidden so that no patient can be identified.
- The indicators can be mapped to the WHO EMT Minimum Data Set (MDS) daily reporting form, which EMTs use to report to the coordination cell [9]. This follows the WHO minimum standards for Emergency Medical Teams [10].
- Power BI row-level security shows each site only its own data. Only M&E and management see all sites.
- Every week the M&E officer checks missing or unusual data and gives feedback to each tent.

## 22.11 Checklist for the IT team

1. Create the Development, Test and Production environments, with Dataverse, in the organisation’s region.
2. Turn on Managed Environments and data loss prevention policies.
3. Create Entra ID groups for each role and site, with MFA and Conditional Access.
4. Enrol devices in Intune and set app protection policies.
5. Set up Dataverse security roles, business units, column security and auditing.
6. Create the SharePoint library and Purview sensitivity labels for patient files.
7. Decide the encryption key (Microsoft-managed or the organisation’s own key).
8. Agree the retention policy with the data protection lead.
9. Build and test the Power BI reports with the M&E team.
10. Run a restore test and a short pilot, including offline use, before go-live.

# 23. References

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